Year 11 Unit 2 Mathematics

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Learning Strategies

Mathematics is often the most challenging subject for students. Much of the trouble comes from the fact that mathematics is about logical thinking, not memorizing rules or remembering formulas. It requires a different style of thinking than other subjects. The students who seem to be "naturally" good at math just happen to adopt the correct strategies of thinking that math requires – often they don't even realise it. We have isolated several key learning strategies used by successful maths students and have made icons to represent them. These icons are distributed throughout the book in order to remind students to adopt these necessary learning strategies:



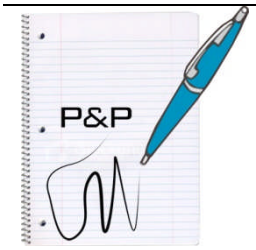
Talk Aloud Many students sit and try to do a problem in complete silence inside their heads. They think that solutions just pop into the heads of 'smart' people. You absolutely must learn to talk aloud and listen to yourself, literally to talk yourself through a problem. Successful students do this without realising. It helps to structure your thoughts while helping your tutor understand the way you think.



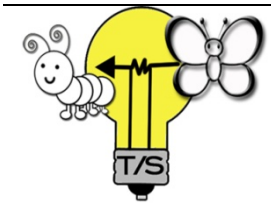
BackChecking This means that you will be doing every step of the question twice, as you work your way through the question to ensure no silly mistakes. For example with this question: $3 \times 2 - 5 \times 7$ you would do "3 times 2 is 5 ... let me check – no 3×2 is 6 ... minus 5 times 7 is minus 35 ... let me check ... minus 5×7 is minus 35. Initially, this may seem time-consuming, but once it is automatic, a great deal of time and marks will be saved.



Avoid Cosmetic Surgery Do not write over old answers since this often results in repeated mistakes or actually erasing the correct answer. When you make mistakes just put one line through the mistake rather than scribbling it out. This helps reduce silly mistakes and makes your work look cleaner and easier to backcheck.



Pen to Paper It is always wise to write things down as you work your way through a problem, in order to keep track of good ideas and to see concepts on paper instead of in your head. This makes it easier to work out the next step in the problem. Harder maths problems cannot be solved in your head alone – put your ideas on paper as soon as you have them – always!



Transfer Skills This strategy is more advanced. It is the skill of making up a simpler question and then transferring those ideas to a more complex question with which you are having difficulty.

For example if you can't remember how to do long addition because you can't recall exactly how to carry the one: $\begin{array}{r} 5889 \\ +\,4587 \\ \hline \end{array}$ then you may want to try adding numbers which you do know how to calculate that also involve carrying the one: $\begin{array}{r} 5 \\ +\,9 \\ \hline \end{array}$

This skill is particularly useful when you can't remember a basic arithmetic or algebraic rule, most of the time you should be able to work it out by creating a simpler version of the question.

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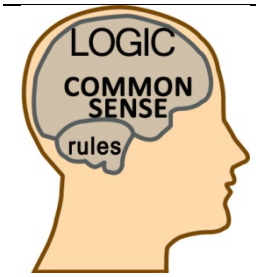
Format Skills These are the skills that keep a question together as an organized whole in terms of your working out on paper. An example of this is using the "=" sign correctly to keep a question lined up properly. In numerical calculations format skills help you to align the numbers correctly.

This skill is important because the correct working out will help you avoid careless mistakes. When your work is jumbled up all over the page it is hard for you to make sense of what belongs with what. Your "silly" mistakes would increase. Format skills also make it a lot easier for you to check over your work and to notice/correct any mistakes.

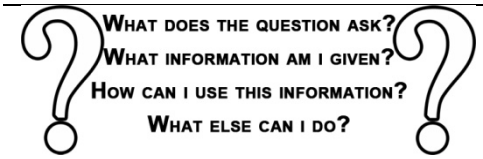
Every topic in math has a way of being written with correct formatting. You will be surprised how much smoother mathematics will be once you learn this skill. Whenever you are unsure you should always ask your tutor or teacher.



Its Ok To Be Wrong Mathematics is in many ways more of a skill than just knowledge. The main skill is problem solving and the only way this can be learned is by thinking hard and making mistakes on the way. As you gain confidence you will naturally worry less about making the mistakes and more about learning from them. Risk trying to solve problems that you are unsure of, this will improve your skill more than anything else. It's ok to be wrong – it is NOT ok to not try.



Avoid Rule Dependency Rules are secondary tools; common sense and logic are primary tools for problem solving and mathematics in general. Ultimately you must understand Why rules work the way they do. Without this you are likely to struggle with tricky problem solving and worded questions. Always rely on your logic and common sense first and on rules second, always ask Why?



Self Questioning This is what strong problem solvers do naturally when they get stuck on a problem or don't know what to do. Ask yourself these questions. They will help to jolt your thinking process; consider just one question at a time and Talk Aloud while putting Pen To Paper.

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Year 11 Unit 2 Mathematics

Basic Arithmetic & Algebra

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Useful formulae and hints

- To add fractions of different denominators, change one or both to equivalent fractions with a common denominator
- To multiply fractions, multiply the denominators, multiply the numerators and simplify if necessary
- To convert fractions to decimals, divide the numerator by the denominator (but learn the simpler conversions by heart)
- To convert fractions to percentages, convert to decimal and then multiply by 100 (but learn the simpler conversions by heart)
- To convert percentages to fractions, remove the percent sign, put the number as the numerator of a fraction with 100 as the denominator, then simplify the fraction if necessary
- To convert decimals to fractions, the numeral(s) after the decimal point form the numerator. The denominator is 10 if the numerator has one digit, 100 if the numerator has 2 digits etc. Example: $0.7 = \frac{7}{10}, 0.41 = \frac{41}{100}, 0.213 = \frac{213}{1000}$. Simplify fraction if necessary
- To convert a recurring decimal, set the recurring part equal to a variable, multiply by 100 and solve
  - $x = 0.\dot{1}\dot{1}$
  - $100x = 11.\dot{1}\dot{1}$
  - $100x = 11 + x$
  - $99x = 11$
  - $x = \frac{11}{99}$
- Distributive law: $(a \times b) + (a \times c) = a(b + c)$
- To rationalize a surd denominator, multiply by its conjugate

  Conjugate of $a + \sqrt{b}$ is $a - \sqrt{b}$

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- When solving inequalities, if we multiply both sides by a negative number, the inequality sign is reversed
- To solve absolute value problems, look at all possible cases: $|x| = 5$ means $x = 5$ or $|x| = -5$
- $a^2 - b^2 = (a - b)(a + b)$

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Exercise 1

Rational Numbers & Surds

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1) Calculate the following, expressing your answers in their simplest form

a) $\frac{3}{4}+\frac{5}{6}$

b) $\frac{2}{11}+\frac{1}{13}$

c) $1\frac{1}{7}+3\frac{3}{5}$

d) $\frac{4}{11}-\frac{4}{22}$

e) $2\frac{2}{3}-3\frac{1}{4}$

f) $\frac{5}{9}-\frac{35}{63}$

2) Simplify the following, expressing your answer in simplest form

a) $\frac{3}{4}\times\frac{8}{9}$

b) $\frac{1}{11}\times\frac{33}{12}$

c) $\frac{7}{2}\times\frac{1}{15}$

d) $\frac{3}{2}\div\frac{9}{10}$

e) $\frac{12}{39}\div\frac{36}{26}$

f) $\frac{1}{2}\div\frac{2}{3}$

3) How many lots of $\frac{3}{10}$ are there in $\frac{2}{5}$

4) Convert the following fractions to decimals

a) $\frac{1}{4}$

b) $\frac{2}{3}$

c) $\frac{1}{8}$

d) $\frac{1}{12}$

5) Convert the following fractions to percentages

a) $\frac{1}{5}$

b) $\frac{3}{4}$

c) $\frac{3}{8}$

d) $\frac{1}{3}$

e) $\frac{7}{1000}$

6) Convert the following percentages to fractions in their simplest form

a) 30%

b) 12.5%

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c) 0.4%

d) 2.5%

7) Convert the following decimals to fractions in their simplest form

a) 0.01

b) 0.4

c) 0.625

d) $0.\dot{1}\dot{5}$

e) Use your result from part d to convert $4.0\dot{1}\dot{5}$ to a mixed numeral

8) Solve or simplify the following by using the distributive law

a) $\left(\frac{3}{5} \times 498\right) + (\frac{3}{5} \times 2)$

b) $\left(\frac{2}{3} \times 2\right) - (\frac{2}{3} \times \frac{1}{2})$

c) $(a+1)(a^2) + (a+1)(2a) + a + 1$

d) $\frac{(a-1)2x+3(a-1)}{a(2x+3)-2x-3}$

9) Convert the following numbers to scientific notation, correct to 3 significant figures

a) 42731

b) 0.91326

c) 6139900

d) 0.034

10) For each of the following numbers, write the number correct to 4 decimal places, and to 4 significant figures

a) 0.043176

b) 0.2565443

c) 0.00012739

d) 1.128755

11) Simplify the following expressions, leaving your answer in surd form

a) $6\sqrt{2} + 2\sqrt{2}$

b) $4\sqrt{8} + 2\sqrt{2}$

c) $\sqrt{27} + 2\sqrt{3}$

d) $2\sqrt{45} + 3\sqrt{20}$

12) Simplify the following, leaving your answer in surd form

a) $\sqrt{108} - \sqrt{48}$

b) $\sqrt{32} - \sqrt{18}$

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c) $3x\sqrt{18} - x\sqrt{128}$

d) $2\sqrt{20} - 4\sqrt{5}$

13) Calculate each of the following leaving your answer in its simplest form

a) $\sqrt{12} \times \sqrt{3}$

b) $\sqrt{3} \times \sqrt{27}$

c) $\sqrt{8} \times \sqrt{50}$

d) $\sqrt{18} \times \sqrt{8}$

e) $\sqrt{1.6} \times \sqrt{5}$

f) $\sqrt{\frac{3}{4}} \times \sqrt{\frac{12}{25}}$

g) $\sqrt{\frac{2}{3}} \times \sqrt{\frac{4.5}{27}}$

14) Evaluate the following by rationalising the denominator, leaving your answers in exact form

a) $\frac{3}{1-\sqrt{2}}$

b) $\frac{\sqrt{6}}{4-\sqrt{6}}$

c) $\frac{32}{\sqrt{2}+1}$

d) $\frac{4a}{\sqrt{3}-1} + \frac{2b}{1-\sqrt{3}}$

15) For what values of a and b is the following expression rational?

$$\frac{a}{2+\sqrt{5}} - \frac{b}{9-4\sqrt{5}}$$

16) Evaluate the following

a) $\sqrt{1.69}$

b) $\sqrt{5\frac{4}{9}}$

c) $\left(4\frac{1}{3}\right)^2$

d) $\sqrt[3]{0.027}$

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Exercise 2

Inequalities & Absolute Values

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1) Solve the following inequalities

a) $x + 2 < 5$

b) $x - 3 > 4$

c) $6t \geq 42$

d) $5y < -30$

e) $\frac{x}{3} > 9$

f) $\frac{2y}{3} \leq 10$

g) $-x < 6$

h) $-2y \leq 3$

2) Solve the following inequalities

a) $2a + 4 \geq 6$

b) $3p - 5 \leq 10$

c) $6 - 3x > 15$

d) $-3 - 4g < 3$

e) $\frac{x}{3} + 4 \leq 2$

f) $\frac{-x}{2} + 10 > 2$

g) $\frac{x}{3} < x + 1$

h) $\frac{-x}{2} > x - 2$

3) Solve the following inequalities

a) $2t + 3 \leq t - 6$

b) $3y - 5 > -y - 4$

c) $2(x - 1) - x < \frac{x}{2}$

d) $3(4 - y) + 2y \geq \frac{1}{2}y$

e) $\frac{x}{2} + \frac{x}{3} \leq 4$

f) $\frac{-x}{3} - \frac{2x}{4} > 6$

g) $\frac{-2p}{3} + \frac{p}{-2} \leq -4$

h) $\frac{-t}{3} - 4 > \frac{t}{2} + 3$

4) Solve

a) $|x| = 3$

b) $|x| = 5$

c) $|x| - 2 = 6$

d) $-|x| = 4$

5) Solve

a) $|x + 2| = 7$

b) $|x - 3| = 4$

c) $|x + 4| = 10$

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d) $-|x-2| = \frac{3}{2}$

6) Solve

a) $|2x-3|-2=6$

b) $|3x+4|-5=\frac{1}{2}$

c) $|2x+1|-4=3$

d) $|x-2|+\frac{1}{2}=\frac{2}{3}$

7) Solve

a) $|x^2+6x-3|=4$

b) $|x^2+2x-1|=2$

c) $|x^2+5x-2|=-6$

d) $|x^2-13|=4$

8) Solve the following algebraically

a) $|x+1|=|x-2|$

b) $\left|x-\frac{1}{2}\right|=|x+3|$

c) $|2x-3|=|x+1|$

d) $|3x+1|=|x+3|$

e) $\left|\frac{1}{2}x+1\right|=|2x-3|+1$

f) $|2x-5|=|x-3|+6$

9) Solve the following graphically

a) $|x-4|=|x+2|$

b) $|2x+2|=|x-1|$

c) $|x-1|=|x+2|+2$

d) $\left|\frac{1}{2}x+1\right|=|x-2|-1$

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Exercise 3

Algebraic Expressions

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1) Simplify the following expressions

a) $3x + 4(x - 2)$

b) $2(3 - y) + 3(2y + 2)$

c) $-(5 - 2t) - (-3 + 4t)$

d) $\frac{1}{2}(4y + 2) - \frac{1}{3}(3 - 6y)$

e) $2 + 5(2y - 4) - 4(4 - y)$

f) $-2 - (-3 - x) + (2x - 3)$

2) Simplify the following expressions

a) $(x^2 - 4x + 3) + (2x^2 + 2x - 6)$

b) $x(2 - x^2) + x^2(2x + 4)$

c) $4(9x - x^2) - 3x(4 - x)$

d) $-2(2x - 2) - 3(-3x - 3)$

3) Simplify the following expressions

a) $(2 + 5y)(2y - 4)$

b) Add $2x + 3$ to $3x - 2$

c) From $\frac{1}{2}x + 4$ subtract $(x - 3)$

d) From $\frac{1}{2}(x + 4)$ subtract $(x - 3)$

e) $2x(3x + 1)(x - 2)$

f) Multiply the sum of $x + 2$ and $x^2 + x$ by $-x$

4) If $x = \sqrt{2}$, evaluate

a) x^2

b) x^8

c) $x^4 - x^2$

d) $\frac{2}{x^2}$

e) $\left(\frac{2}{x}\right)^2$

5) If $x = \frac{a^2 b}{c^2}$, calculate the value of x when

a) $a = 1, b = 2, c = 3$

b) $a = (\frac{1}{5})^2$, $b = \left(\frac{2}{5}\right)^{\frac{1}{2}}$, $c = \sqrt[4]{2}$

c) $a = \frac{1}{2b}$, $c = 2\sqrt{b}$

d) $a = \frac{1}{2b}$, $c = \frac{1}{2\sqrt{b}}$

6) The area of a circle is given by the formula, $A = \pi r^2$; calculate the radius of the circle (to 2 d.p.)when its area is 12 cm^2

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7) The kinetic energy of an object can be calculated from the formula:$KE = \frac{1}{2}mv^2$, where m is the mass of the object (in kilograms) and v is its velocity (in meters per second). Calculate the kinetic energy of an object in each of the cases below

a) Mass of 2kg and a velocity of 4 meters per second

b) Mass of 500 grams and a velocity of 10 meters per second

c) Mass of 10kg and a velocity of 10 kilometres per second

d) Mass of 250 grams and a velocity of 24000 centimetres per minute

8) The volume of a cone is given by the formula $V = \frac{1}{3}\pi r^2 h$. What is the radius of a cone of volume 1200 $cm^3$and height 100cm?

9) If a set of three resistors is connected in parallel, the equivalent resistance (R) of the set is given by the formula $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$. Calculate the resistance of the set (in ohms) if:

a) $R_1 = R_2 = R_3 = 2\ ohms$

b) $R_1 = 2\ ohms, R_2 = 3\ ohms, R_3 = 4\ ohms$

c) $R_1 = 0.5\ ohms, R_2 = 2\ ohms, R_3 = 0.25\ ohms$

d) $R_1 = \frac{1}{2}R_2, R_2 = \frac{1}{2}R_3$ (express your answer in terms of R_3)

10) Simplify the following by removing the common factor

a) $4p^2 - 2p$

b) $3y^3 + 2y$

c) $6x^2 + 4x + 2$

d) $10x^3 - 4x^2 + 8x$

e) $-4x - 3x^2$

f) $\frac{1}{x} + \frac{1}{x^2}$

11) Simplify the following expressions involving the difference of two squares

a) $x^2 - 4$

b) $4y^2 - 9$

c) $25p^2 - 25$

d) $\frac{x^2}{4} - \frac{1}{16}$

e) $y^4 - 100y^2$

f) $x^2 - 2$

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12) Factorise the following

a) $x^2 - 6x + 9$

b) $x^2 + 4x - 5$

c) $x^2 - 8x + 12$

d) $2x^2 + 9x + 10$

e) $3x^2 + 5x - 12$

f) $6x^2 - 14x + 8$

13) Factorise the following

a) $bx + by - 3x - 3y$

b) $x^2 + xy - \frac{1}{4} - \frac{1}{2}y$

c) $4x^2 - y^2 + 8x - 4y$

d) $x^3 + 1$

e) $x^3 - 27$

f) $x^3 + 125$

14) Reduce the following fractions to their simplest form

a) $\frac{2p+3t}{8p+12t}$

b) $\frac{3x-y}{4y-12x}$

c) $\frac{1-x}{1-x^2}$

d) $\frac{x^2+5x+6}{x+3}$

e) $\frac{y^2-16}{y-4}$

f) $\frac{x^2+9}{x-3}$

15) Simplify

a) $\frac{x}{x-2} \times \frac{x^2-4}{xy+x}$

b) $\frac{2}{x+1} \times \frac{(x+1)^2}{4}$

c) $\frac{x^2-8x+7}{t^2-6t+9} \times \frac{t-3}{x-7}$

d) $\frac{2}{p-2} \div \frac{4p}{p^2-4}$

e) $\frac{3x}{2x-1} \div \frac{6x}{4x-2}$

f) $\frac{xy}{x+1} \div \frac{x^2y^2}{4x+4}$

16) Simplify

a) $\frac{a+b}{3} + \frac{a-b}{9}$

b) $\frac{2x+3y}{2} - \frac{2y-3x}{3}$

c) $\frac{2-p}{p} + \frac{p+2}{2}$

d) $\frac{4}{y} - \frac{6}{y(y-1)}$

e) $\frac{1}{x^2+5x+4} + \frac{2}{x+1}$

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f) $\frac{1}{x^2-4}+\frac{1}{x^2+4x+4}$

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Exercise 4

Linear & Quadratic Expressions

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1) Solve the following linear equations

a) $2x + 4 = 10$

b) $3x + 7 = 4$

c) $\frac{1}{2}x - 4 = 5$

d) $\frac{2}{3}x + 6 = 8$

e) $2 - \frac{4}{5}x = -6$

f) $11 - \frac{1}{2}x = 11$

2) Solve the following linear equations

a) $\frac{4x+1}{x} = 3$

b) $\frac{2x-6}{x} = -4$

c) $\frac{4x-2}{x-2} = 8$

d) $\frac{3x+6}{x-4} = 10$

e) $\frac{\frac{1}{2}x-4}{x+1} = 6$

f) $\frac{2-\frac{1}{3}x}{x-3} = 3$

3) Solve the following linear equations

a) $\frac{x-4}{x+2} = \frac{x-3}{x+1}$

b) $\frac{x+1}{x-1} = \frac{x+2}{x-6}$

c) $\frac{2x-3}{3x-2} = \frac{2x+1}{3x-1}$

d) $\frac{\sqrt{x}+2}{\sqrt{x}-1} = \frac{\sqrt{x}-4}{\sqrt{x}+1}$

4) Find the values of x for which

a) $2x + 2 > 6$

b) $4x - 3 \leq 9$

c) $2 - 6x \geq -10$

d) $1 - \frac{1}{2}x < 3$

e) $|x - 2| < 5$

f) $|x + 1| \geq 3$

g) $|x + 1| + 1 < 2$

5) Solve the following equations by factorization

a) $x^2 + 5x - 6 = 0$

b) $x^2 - 5x + 6 = 0$

c) $x^2 + 2x + 1 = 0$

d) $2x^2 + 7x - 9 = 0$

e) $6x^2 - 14x + 8 = 0$

f) $10x^2 + 6x - 4 = 0$

g) $10x^2 - 6x - 4 = 0$

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6) Solve the following equations using the most appropriate method

a) $6 - x^2 = x$

b) $8x^2 + 2x - 1 = 0$

c) $x^2 = 8x$

d) $(x - 4)^2 = 9$

e) $2x^2 + 4x + 4 = 0$

f) $x^2 = 4x - 2$

7) Solve the following simultaneous equations. Check your results by substitution into the original equations

a) $2x + 3y = 5$ and
$x + y = 2$

b) $x - 4y = -10$ and
$x - y = -1$

c) $x + \frac{3}{2}y = -\frac{5}{2}$ and
$2x - y = 3$

d) $4x - 2y = 3$ and
$x + y = 0$

e) $x - y = 4$ and $-x + y = 8$

f) $x - y = 2$ and
$-x + y = -2$

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Year 11 Unit 2 Mathematics

Real Functions

Useful formulae and hints

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- The domain of a function is the set of all values of x for which the values of the function are real
- The range of a function is the set of all y values that result from applying the function rule to all x values in the domain
- A function can have only one y value for each x value in the domain
- The x intercepts of a function are the values (if any) at which the function equals zero
- The y intercept of a function is the value of the function when $x = 0$
- An asymptote is a value that a curve approaches but never reaches
- A discontinuity is a point where a function is not defined
- The general equation of a circle is $(x - h)^2 + (y - k)^2 = r^2$, where h and k are the co-ordinates of the centre, and r is the radius
- The general equation of a parabola is: $(y - k)^2 = 4A(x - h)$, where h and k are the co-ordinates of the vertex. The vertical (or horizontal) distance from the vertex to the focus, and from the vertex to the directrix is A. The focus lies within the parabola, the directrix is a line that lies outside the parabola

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Exercise 1

Range, Domain & Variables

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1) State the domain and range (from the set of real numbers) of the following functions

a) $f(x) = x^2 - 1$

b) $f(x) = \sqrt{x}$

c) $f(x) = \frac{1}{x}$

d) $f(x) = \frac{1}{\sqrt{x}}$

e) $f(x) = x + 1$

f) $f(x) = \sqrt{-x}$

2) Find the range and domain of the following functions

a) $f(x) = \sqrt{x} + 2$

b) $f(x) = \sqrt{x} + 1$

c) $f(x) = \sqrt{x} - 2$

d) $f(x) = \sqrt{x} + c$

3) Find the range and domain of the following functions

a) $f(x) = x^2$

b) $f(x) = x^2 + 1$

c) $f(x) = x^2 - 2$

d) $f(x) = x^2 + c$

4) Find the range and domain of the following functions

a) $f(x) = (x + 1)^2$

b) $f(x) = (x - 2)^2$

c) $f(x) = (x + 4)^2$

d) $f(x) = (x + c)^2$

5) Which of the following are not functions; give reasons for those considered non-functions

a) $f(x) = \frac{1}{\sqrt{x}}$

b) $f(x) = 2$

c) $f(x)^2 - 2x = 3$

d) $f(x^2) - 2x = 3$

e) $x = -2$

f) $x^2 + y^2 = 4$

6) Find the range and domain of the following functions

a) $y = \frac{1}{x}$

b) $y = \frac{1}{x} + 1$

c) $y = \frac{1}{x} - 1$

d) $y = \frac{1}{x} + c$

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7) Find the range and domain of the following functions

a) $y = \frac{1}{x^2}$

b) $y = \frac{1}{x^2+1}$

c) $y = \frac{1}{x^2-1}$

d) $y = \frac{1}{x^2+c}$

8) Find the range and domain of the following functions

a) $y = |x|$

b) $y = |x + 1|$

c) $y = |x - 2|$

d) $y = |x + c|$

e) $y = |x| + 1$

f) $y = |x| - 2$

g) $y = |x| + c$

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Exercise 2

Properties of Graphs of Real Functions

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For each question below, sketch the graph of the function, and determine the following properties.

- x intercept
- y intercept
- Where the function is increasing
- Where the function is decreasing
- Where the function is positive, negative, and zero
- Any horizontal or vertical asymptotes
- The maximum and minimum values of the function
- If there are any discontinuities

Use the last equation in each question to generalize the above properties of functions of that type

1) Linear functions

a) $y = 2x$

b) $y = 3x + 1$

c) $y = 4x - 2$

d) $y = ax + b$

2) Quadratic functions

a) $y = x^2$

b) $y = x^2 + 1$

c) $y = x^2 - 2$

d) $y = x^2 + c$

3) Inverse functions

a) $y = \frac{1}{x}$

b) $y = \frac{1}{x+1}$

c) $y = \frac{1}{x-2}$

d) $y = \frac{1}{x} + 1$

e) $y = \frac{1}{x} - 2$

f) $y = \frac{1}{x+a} + b$

4) Radicals

a) $y = \sqrt{x}$

b) $y = \sqrt{x+1}$

c) $y = \sqrt{x-2}$

d) $y = \sqrt{x} + 1$

e) $y = \sqrt{x} - 2$

f) $y = \sqrt{x+a}$

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g) $y = \sqrt{x} + b$

5) Absolute value

a) $y = |x|$

b) $y = |x + 1|$

c) $y = |x - 2|$

d) $y = |x| + 1$

e) $y = |x| - 2$

f) $y = |x + a|$

g) $y = |x| + b$

6) Miscellaneous functions

a) $y = x \; for \; k \leq x < k + 1, for \; all \; integers \; k$

b) $y = \frac{1}{x^2}$

c) $y = \frac{1}{\sqrt{x}}$

d) $y = \frac{1}{|x|}$

e) $y = x^3 + c$

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Exercise 3

Geometric Representation

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1) Write the equation of the following circles

a) Centre at the origin, radius of 1 units

b) Centre at the origin, radius 2 units

c) Centre at the point (0,1), radius 2 units

d) Centre at point (1,-1), radius 3 units

e) Centre at point (2,3), radius 4 units

f) Centre at point (½, ½), radius 1.5 units

2) Describe the circle given by the following equations

a) $x^2 + y^2 = 9$

b) $x^2 + y^2 - 4y + 2 = 0$

c) $x^2 + y^2 - 2x - 2y - \frac{3}{2} = 0$

d) $x^2 + 4x + y^2 - 4y = 2$

e) $x^2 + 6x + y^2 + 2y + 9 = 0$

f) $x^2 + x + y^2 + y = 0$

3) Determine the vertex and focus of the following parabolas

a) $y = \frac{x^2}{4}$

b) $y = 2x^2 - 4x + 4$

c) $6y = x^2 + 4x + 16$

d) $16y = x^2 + 6x + 73$

e) $8x = -y^2 + 4y - 12$

f) $26y = -x^2 - x + \frac{155}{4}$

4) Find the equation of the parabola that has:

a) Vertex at $(-1, 3)$, focus at (-1, -3)

b) Vertex at $\left(0, -\frac{1}{2}\right)$, focus at (0, 4)

c) Vertex at (3, -1), focus at (3, 5)

d) Vertex at $\left(-\frac{3}{4}, \frac{1}{4}\right)$, focus at $\left(-\frac{3}{4}, 0\right)$

e) Vertex at (0, 0), focus at (0, 1.5)

f) Vertex at (0, -1), focus at (2, -1)

5) Find the equation of the parabola that has

a) Vertex at (0, 0), directrix $y = 2$

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b) Vertex at (-1, 2), directrix $y = -3$

c) Vertex at $\left(\frac{1}{2}, 1\right)$, directrix $y = 4$

d) Vertex at (1, 1) directrix $x = 2$

e) Vertex at $\left(\frac{3}{4}, -\frac{1}{2}\right)$, directrix $y = 3$

f) Vertex at (3, 2), directrix $x = 0$

6) Find the equation of the parabola that has

a) Focus at (0, 0), directrix $y = -2$

b) Focus at (2, -2), directrix $y = \frac{3}{2}$

c) Focus at $\left(\frac{1}{2}, -\frac{1}{2}\right)$, directrix $y = \frac{7}{2}$

d) Focus at (1, 1), directrix $x = 3$

e) Focus at $\left(2, \frac{3}{4}\right)$, directrix $y = -\frac{7}{4}$

f) Focus at (-2, 3), directrix $x = -5$

7) Sketch the following curves, showing centre and radius for circles; and focus, directrix and vertex for parabolas

a) $x^2 + y^2 = 16$

b) $y = 4x^2 - 2x + 8$

c) $x^2 + y^2 + 4x - 6y + 10$

d) $y = 10x^2 - 6x + 3$

e) $y^2 + 2 = 2x - 4 + 6y$

f) $2y^2 + 8y - 4x = x^2 - 2$

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Exercise 4

Graphing Inequalities

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1) Sketch and label the region bounded by

a) The x axis, the y axis, and the inequality $y < -2x + 3$

b) The x axis and the inequalities $y < -x + 2$ and $y < x + 2$

c) The inequality $y > -\frac{1}{2}x + 4$

d) The inequalities $y < 4$ and $y > 0$

e) The inequalities $|x| < 2$ and $y < \frac{1}{2}x + 1$

f) The inequality $y > 3x - 3$

2) Sketch and label the region bounded by

a) The inequalities $y > x^2$ and $y < 1$

b) The inequalities $x > 0$, $y > 0$, and $y < -x^2$

c) The inequalities $y < 0$ and $y > x^2 + 4x + 3$

d) The x axis, and the inequalities $y < x^3$ and $x < 2$

e) The inequalities $x > 0$, $y > 0$, and $y < \frac{1}{x}$

f) The inequalities $x > 0$, $y > \sqrt{x}$, and $y < -x + 4$

3) Sketch and label the region bounded by

a) The inequalities $x^2 + y^2 < 1$, $x > 0$, and $y > 0$

b) The inequalities $x^2 - 4x + y^2 < 0$ and $y > 1$

c) The inequalities $x^2 + y^2 < 4$ and $y > -x$

d) The inequalities $x^2 + y^2 + 2x - 2y > 7$, $x > -4$, and $y < 4$

4) Find a system of inequalities whose solutions correspond to the regions described; sketch the regions

a) The points lying inside the circle with centre (1, 1) and radius 2, but to the right of the line $x = 2$

b) The points whose boundary consists of portions of the x axis, the ordinates at $x = 2$, $x = 3$, and the curve having its turning point at $\left(\frac{5}{2}, 4\right)$, which is also its maximum

c) The points where y is greater than x and both x and y are negative

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d) The triangle bounded by the points (0, 2), (1, 1) and the origin

e) The region inside the circle of radius 2, centred at (2, 1) and the points for which y is greater than 1. Describe the shape formed

f) The region inside the circle of centre (-2, 4) with radius 1, and the points for which x is greater than -1

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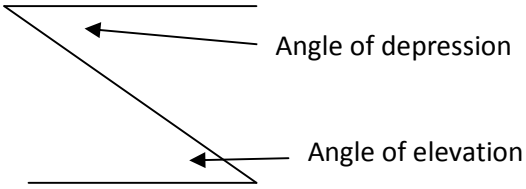
Year 11 Unit 2 Mathematics

Basic Trigonometry

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Useful formulae and hints

- $\sin x$ is the vertical distance of the point from the origin
- $\cos x$ is the horizontal distance of the point from the origin
- Bearings are measured from North in a clockwise direction
- Angle of elevation is measured from the ground looking up and is equal to the angle of depression.



- Sine rule: $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$, where A, B, C are the angles opposite sides a, b, c respectively
- Cosine rule: $a^2 = b^2 + c^2 - 2bc\cos A$
- Area of a non-right angled triangle is $\frac{1}{2}ab\sin C$

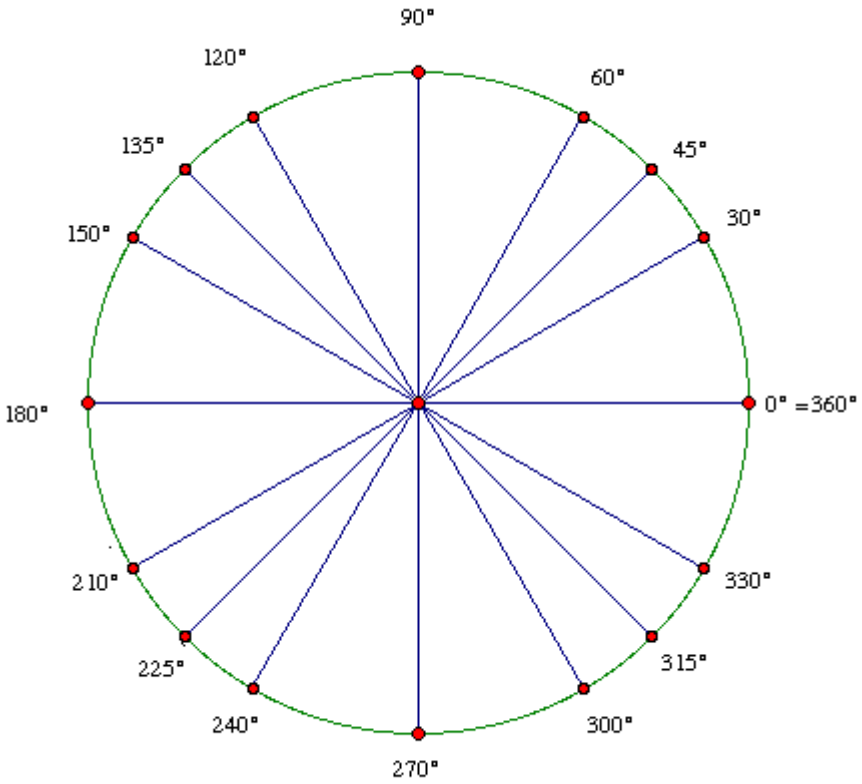
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Exercise 1

Trigonometric Ratios and Identities

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1) For each point on the unit circle write a co-ordinate pair that represents $(\cos x\,, \sin x)$, where x is the angle measurement shown on the appropriate point



2) Complete the following definitions in terms of $\sin x$ and $\cos x$

a) $\tan x =$ _____

b) $\csc x =$ _____

c) $\sec x =$ _____

d) $\cot x =$ _____

3) For what values of θ are the above trigonometric ratios not defined?

4) Graph the following

a) $\sin x$ for x between 0 and 360°

b) $\tan x$ for x between 0 and 360°

c) $\sec x$ for x between 0 and 360°

5) Complete the following in terms of θ

a) $\sin(-\theta) =$ _____

b) $\cos(90° - \theta) =$ _____

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c) $\tan(180° + \theta) =$ _____

d) $\csc(90° - \theta) =$ _____

e) $\sec(-\theta) =$ _____

6) Complete the following trigonometric identities

a) $\sin^2\theta + \cos^2\theta =$ _____

b) $1 + \tan^2\theta =$ _____

c) $1 + \cot^2\theta =$ _____

d) $\sin(2\theta) =$ _____

e) $\cos(2\theta) =$ _____

7) Solve the following, showing all possible solutions in the domain

a) $4\cos x = 1 + 2\cos x$, for $0° \leq x \leq 90°$

b) $\csc^2 x = 2$, for $0° \leq x \leq 180°$

c) $4\sin x = 1 + 2\sin x$, for $-90° \leq x \leq 90°$

d) $\cot x = 2\cos x$, for $-180° \leq x \leq 180°$

e) $10\cos 2x = 4\cos 60°$, for $0° \leq x \leq 360°$

f) $\cot^2 x = \csc^2 x$, for $0° \leq x \leq 90°$

g) $2\sin x - \sin 30° = \cos 0°$, for $90° \leq x \leq 180°$

8) Using exact values, simplify the following: leave answer in surd form if necessary

a) $\cos 30° \tan 30°$

b) $\sec 45° - \sin 45°$

c) $\csc 60° \sec 30°$

d) $\dfrac{\sec 30° \tan 60°}{\csc^2 45°}$

e) $(\tan 30° + \csc 60°)\cos 30°$

f) $\sin^2 27° + \left(\dfrac{1}{\sec 27°} \times \cos 27°\right)$

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Exercise 2

Angles of Elevation & Bearings

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1) Sketch and label the following bearings

a) 030°

b) 075°

c) 120°

d) 135°

e) 180°

f) 240°

g) 280°

h) 300°

i) 345°

2) Sketch the following directions and write their bearings

a) Due South

b) South-East

c) North-West

d) North-East

e) Due North

3) Sketch diagrams that show the following

a) A man travels due East for x km then due South for y km

b) A man travels North-East for x km, then due South for y km

c) A man travels on a bearing of 45° for x km , then on a bearing of 225° for y km

d) A man travels on a bearing of 330° for x km, then on a bearing of 210° for y km

e) A man travels due South for x km, then travels due East for y km, he then walks back to his starting point for z km.

4) Solve the following (the diagrams from Q3 may be useful)

a) A man travels due East for 3 km, then travels due South for 4 km. What is the shortest distance back to his original starting position?

b) A man travels North-East then turns and travels due South for 15 km until he is due East of his starting position. How far due East of his starting position is he?

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c) A man travels on a bearing of 45° for 10 km; he then travels on a bearing of 225° for 12 km. What is the shortest distance back to his original starting position?

d) A man travels on a bearing of 330° for 4 km and then on a bearing of 210° for 4 km. How far and on what bearing is his shortest path back to his original starting position?

e) A man travels due South for 6 km, then due East for 6 km. On what bearing must he travel and for what distance to take the shortest path back to his starting position?

5) Solve the following

a) Two friends Bill and Ben leave from the same point at the same time. Bill walks North-East at 4 km per hour for 2 hours. Ben walks at a rate of 3 km per hour for 2 hours South-East. How far apart are they at this time?

b) Fred travels due East then walks on a bearing of 300° for 8 km until he is due North of his original starting position. How far away from his original position is he? How far due East did he walk?

c) Alan and Ken each start rowing a boat from the same position. Alan rows due west for 10 km, whilst Ken rows for 20km at which time he is directly South of Alan. On what bearing did Ken row, and what distance was he away from Alan when he was due south of him?

6) Solve the following

a) A 3 meter ladder leans against a wall and makes an angle of 50° with the ground. How high up the wall does the ladder reach?

b) The light from a tower shines on an object on the ground. The angle of depression of the light is 75°. If the tower is 20 metres high, how far away is the object from the base of the tower?

c) A 4 meter pole casts a 10 metre shadow. What is the angle of elevation of the pole from the end of the shadow?

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d) From the top of a cliff the angle of depression to a boat on the ocean is 2°. If the cliff is 100 metres high, how far out to sea is the boat?

e) A fire fighter has to use his 20 metre ladder to reach the window of a burning apartment building. If the window is 15 meters from the ground, on what angle would the ladder be placed so it can be reached?

f) A peg on the ground sits between two poles. The first pole is 2 metres high and the other is 7.66 metres high. From the peg a rope of length 4 metres is attached to the top of the first pole. Another rope of length 10 metres is attached to the top of the second pole. What angle is made between the two ropes?

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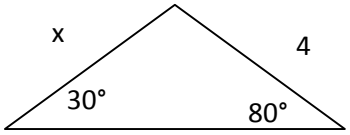
Exercise 3

Non-right Angled Triangles

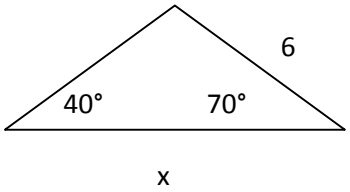
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1) Solve the following using the sine rule. Note for questions where the angle is unknown, round your answer to one decimal place, and ensure all possible solutions are found. (Diagrams are not drawn to scale)

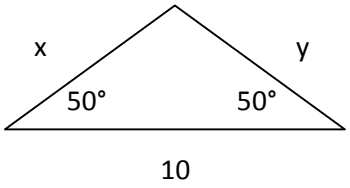
a)



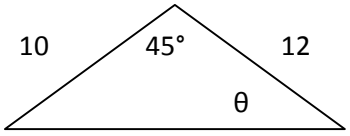
b)



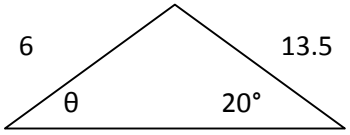
c)



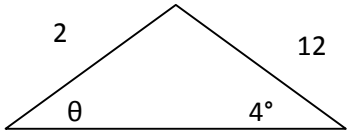
d)



e)



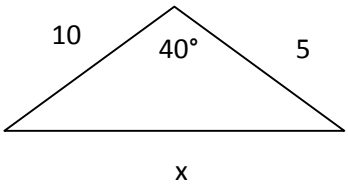
f)



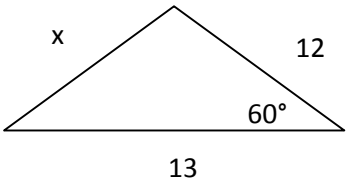
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2) Solve the following using the cosine rule. Note for questions where the angle is unknown, round your answer to one decimal place, and ensure all possible solutions are found. (Diagrams are not drawn to scale)

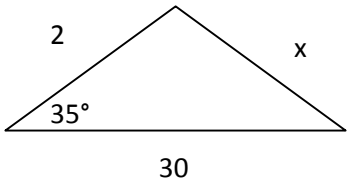
a)



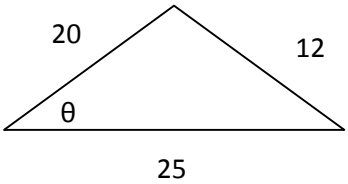
b)



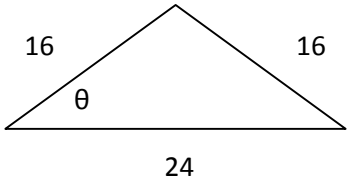
c)



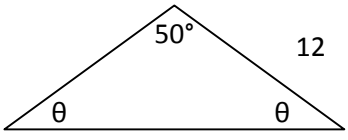
d)



e)



f)



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3) Find the area of the triangles in question 2 by using the sine formula

4) Solve the following by using the sine rule or cosine rule; draw a diagram to help solve

a) A post has been hit by a truck and is leaning so it makes an angle of 85° with the ground. A surveyor walks 20 metres from the base of the pole and measures the angle of elevation to the top as 40°. How tall is the pole if it is leaning toward him? How tall is the pole if it is leaning away from him?

b) Boat A travels due east for 6 km. Boat B travels on a bearing of 130° for 8 km. How far apart are the boats?

c) A mark is made on the side of a wall. A man 40 metres from the base of the wall measures the angle of elevation to the mark as 20°, and the angle of elevation to the top of the wall as 60°. How far is the mark from the top of the wall?

d) What is the perimeter of a triangle with two adjacent sides that measure 15 and 18 metres respectively, with the angle between them 75°?

e) The pilot of a helicopter hovering above the ocean measures the angle of depression to ship A on its left at 50°, and the angle of depression to ship B on its right at 70°. If the ships are 200 metres apart, how high above the ocean is the helicopter hovering?

f) A car travels 40 km on a bearing of 70°; then travels on a bearing of 130° until it is exactly due east of its starting position . What is the shortest distance back to its starting position?

5) Find the areas of the triangles used in question 4 parts a, b and d

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Year 11 Unit 2 Mathematics

Lines & Linear Functions

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Useful formulae and hints

- The roots of an equation is/are the point(s) where the equation equals zero
- Parallel lines have the same gradient
- If the gradient of a line is a, the gradient of a line perpendicular is $-\frac{1}{a}$
- The general equation of a line is $y = mx + c$, where m is the gradient and c is the y-intercept
- If lines do not have the same gradient they must intersect at a point
- If two equations have the same gradient and pass through the same point, the equations represent the same line
- The distance between two points $(x_1, x_2)$and(y_1, y_2) is $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
- The midpoint between two points $(x_1, x_2)$and(y_1, y_2) is

$$m = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

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Exercise 1

Algebraic Properties of Lines

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1) What is the root of each of the following linear equations?

a) $2x - 4 = 0$

b) $3x - 3 = 0$

c) $4x - 2 = 0$

d) $x + 5 = 0$

e) $4x + 2 = 0$

f) $3x + 1 = 0$

g) $x + \frac{1}{2} = 0$

h) $2x - \frac{1}{8} = 0$

i) $2x + 4 = 6$

j) $3x - 2 = -3$

2) Each equation in column 1 is parallel to one of the lines in column 2. Match the parallel lines

Column 1	Column 2
$y = 3x + 1$	$y = -x + 9$
$y = \frac{1}{2}x + 2$	$4y = 6x - 1$
$-4y = 2x - 1$	$-\frac{5}{2}y = -\frac{5}{4}x$
$-2y = 2x + 3$	$y = 3x + 10$
$-\frac{1}{2}y = x - 6$	$-3y = 6x - 4$
$\frac{1}{3}y = -\frac{1}{2}x + 4$	$6y = -3x + 2$

3) Each equation in column 1 is perpendicular to one of the lines in column 2. Match the perpendicular lines

Column 1	Column 2
$y = x + 4$	$2y = -x + 3$
$y = 2x - 2$	$y = \frac{1}{2}x - 3$
$2y = 3x + 1$	$y = -x - 3$
$\frac{1}{2}y = 2x$	$3y = 3x + 2$
$-3y = 6x + 2$	$-4y = x - 5$
$-y = x - 8$	$3y = -\frac{2}{3}x + 2$

4) Write the equation of the following lines

a) Having a slope of 1 and passing through the point (2,4)

b) Having a slope of 2 and passing through the point (0,2)

c) Having a slope of 4 and passing through the point (-2,-1)

d) Having a slope of -1 and passing through the point (3,1)

e) Having a slope of -2 and passing through the point (2,2)

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f) Having a slope of -2 and passing through the point (-1,-3)

g) Having a slope of ½ and passing through the point (1,0)

h) Having a slope of $\frac{2}{3}$ and passing through the point (1,3)

i) Having a slope of $-\frac{1}{2}$ and passing through the point (2,1)

j) Having a slope of $-\frac{3}{4}$ and passing through the point (3,0)

k) Having a slope of $-\frac{2}{3}$ and passing through the point (-3,-2)

5) Write the equation of the lines passing through the following pairs of points

a) (1,1) and (2,2)

b) (1,4) and (3,6)

c) (2,0) and (4,4)

d) (-1,3) and (-3,6)

e) (2,-1) and (-2,5)

f) (-3,-3) and (0,-1)

g) $(\frac{1}{2}, 2)$ and $(-\frac{1}{2}, 4)$

h) (-2,-6) and (-1,11)

6) Find the equation of the following lines

a) Parallel to the line $y = 2x + 1$ and passing through the point (1,1)

b) Parallel to the line $y = x - 4$ and passing through the point (0,3)

c) Parallel to the line $2y = 3x + 1$ and passing through the point (-2,4)

d) Parallel to the line $\frac{1}{2}y = x - 2$ and passing through the point (2,0)

e) Parallel to the line $3x - 2y + 4 = 0$ and passing through the point (-1,-2)

f) Parallel to the line $x + 4y - 2 = 0$ and passing through the point (-2,0)

g) Parallel to the line $2x - 2y + 3 = 0$ and passing through the point $(-\frac{1}{2}, -\frac{1}{2})$

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7) Find the equation of the following lines

a) Perpendicular to the line $y = -\frac{1}{2}x + 1$ and passing through the point (0,0)

b) Perpendicular to the line $y = -\frac{1}{4}x - 2$ and passing through the point (1,-1)

c) Perpendicular to the line $y = 2x + 4$ and passing through the point (-2,-1)

d) Perpendicular to the line $2y = -x - 3$ and passing through the point (3,1)

e) Perpendicular to the line $3x - 2y + 1 = 0$ and passing through the point (2,0)

f) Perpendicular to the line $4x + y + 2 = 0$ and passing through the point (-1, $\frac{1}{2}$)

g) Perpendicular to the line $y = x$ and passing through the point (3,1)

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Exercise 2

Intersection of Lines

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1) Which of the following pairs of lines intersect? Give your reasons.

a) $2x + 3y + 2 = 0$ and $y = -\frac{3}{2}x - 2$

b) $y - 2 = 2x + 4$ and $y = 2x$

c) $y - x = 0$ and $y + x = 0$

d) $2y + 4 - 3x = -x + 5$ and $x - y = 0$

e) $\frac{1}{2}x - 4y + 3 = -\frac{7}{2}y$ and $-x - 4y + 3 = \frac{7}{2}y$

f) $\sqrt{2}y = \sqrt{x}$ and $y = \frac{\sqrt{2x}}{2}$

2) Give example equations of each of the following pairs of lines

a) Two lines that intersect at a point

b) Two lines that intersect at an infinite number of points

c) Two lines that intersect at two points

d) Two lines that never intersect

3) At what point(s) do the following pairs of lines intersect?

a) $y = x + 2$ and $2y = x - 4$

b) $2y - x + 4 = 0$ and $x + 4y + 2 = 0$

c) $\frac{1}{2}x + 3 = y - 3$ and $y = x + 1$

d) $2y = 4x - 6$ and $3y - 6x + 9 = 0$

e) $2x - y + 1 = 0$ and $y - 3x = 4$

f) $y = \frac{1}{2}x + 5$ and $2y - x + 4 = 0$

g) $y = x$ and $-x = y$

h) $x - 2 = y + 3x$ and $\frac{1}{2}y - \frac{5}{2} = 6x$

4) Find the equation of the following

a) The line that has a slope of -2, and passes through the point of intersection of the lines $y = 2x - 1$ and $y = 3x - 2$

b) The line that passes through the origin, and also passes through the intersection of the lines $2x - y = 2$ and $\frac{1}{2}y + 1 = x$

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c) The line that passes through the intersection of the lines $2y + x = 5$ and $-x + y = 4$, and is also perpendicular to the second line

d) The line that passes through the point (-2,-1) and also passes through the intersection of the lines $y = \frac{1}{2}x + 2$ and $\frac{1}{2}y - x = -1$

e) The line that passes through the intersection of $y = 2x$ and $y = -3x + 5$, and is also parallel to the first line

5) Shade the region(s) of the number plane as defined in the following questions

a) The region where $y < 1 - x$ and $2y > x + 2$

b) The region where $y > x + 2$ and $-\frac{1}{2}y > \frac{1}{2}x + 4$

c) The region where $2x + y < 4$ and $x - 2y < -3$

d) The region where$(-x - y) > 0$ and $\frac{1}{2}x < -(y + 1)$

6) Draw and describe

a) The region bounded by the inequalities $2y < 3x - 1,\ -y > 2x - 10$ and $3y > x + 2$

b) The equations of the lines that pass through each of the following pairs of points

i. (-2,1) and (0,0)

ii. (-4,-4) and (-2,1)

iii. (-4,-4) and (0,0)

c) The inequalities that form a triangle bounded by the lines in part b

d) Show in your diagram and by substitution into the inequalities that the point (3,2) lies within the triangle.

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Exercise 3

Distance & Midpoints

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1) Find the distance between the following pairs of points. Leave answer in surd form if necessary.

a) (2,2) and (1,1)

b) (3,4) and (0,2)

c) (2,6) and (1,3)

d) (1,4) and (3,3)

e) (0,2) and (2,1)

f) (4,5) and (6,2)

2) Find the distance between the following pairs of points. Leave answer in surd form if necessary

a) (-3,-1) and (1,-2)

b) (0,-3) and (-2,1)

c) (-1,-2) and (3,-4)

d) (4,-1) and (0,-3)

e) (2,2) and (-1,1)

f) (1,1) and (-3,3)

3) Find the distance between the following points. Leave answer in surd form if necessary

a) $(\frac{1}{2},\frac{1}{2})$ and $(\frac{3}{2},0)$

b) $(\frac{5}{2},\frac{3}{2})$ and $(6,\frac{1}{2})$

c) $(0,-\frac{1}{2})$ and $(\frac{3}{2},4)$

d) $(-\frac{3}{2},\frac{1}{2})$ and (2, -2)

e) $(\frac{1}{2},-\frac{1}{2})$ and $(-\frac{1}{2},\frac{1}{2})$

f) $(\frac{3}{2},\frac{1}{2})$ and $(-\frac{3}{2},-\frac{1}{2})$

4) Find the midpoints of the line segments joining the following pairs of points

a) (2,2) and (1,1)

b) (3,4) and (0,2)

c) (2,6) and (1,3)

d) (1,4) and (3,3)

e) (0,2) and (2,1)

f) (4,5) and (6,2)

5) Find the midpoints of the line segments joining the following pairs of points

a) (-3,-1) and (1,-2)

b) (0,-3) and (-2,1)

c) (-1,-2) and (3,-4)

d) (4,-1) and (0,-3)

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e) (2,2) and (-1,1)

f) (1,1) and (-3,3)

6) Find the midpoints of the line segments joining the following pairs of points

a) $(\frac{1}{2},\frac{1}{2})$ and $(\frac{3}{2},0)$

b) $(\frac{5}{2},\frac{3}{2})$ and $(6,\frac{1}{2})$

c) $(0,-\frac{1}{2})$ and $(\frac{3}{2},4)$

d) $(-\frac{3}{2},\frac{1}{2})$ and (2, -2)

e) $(\frac{1}{2},-\frac{1}{2})$ and $(-\frac{1}{2},\frac{1}{2})$

f) $(\frac{3}{2},\frac{1}{2})$ and $(-\frac{3}{2},-\frac{1}{2})$

7) Find the perpendicular distance from each line to the point given

a) $y = 2x + 2$ and the point (1,2)

b) $3y - x = 1$ and the point (-1,3)

c) $y = -x$ and the point (2,0)

d) $2y + x - 2 = 0$ and the point (-2,1)

e) $\frac{1}{2}y = x - 2$ and the point (1,-1)

f) $y = 4$ and the point (2,4)

8) Draw the line segment (A) connecting the points (1, 2) and (3, 8). Also draw the line segment (B) connecting the points (-2,-10) and (1,-1). Find the midpoint of each line segment, the length of each line segment, and the equation of the line joining the midpoint of A to the midpoint of B.

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Year 11 Unit 2 Mathematics

Quadratic Polynomials

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Useful formulae and hints

- Completing the square puts an equation into the form $y = (x + a)^2 + b$
- The determinant of a function of the form $y = ax^2 + bx + c$ is

  Det $= b^2 - 4ac$

- The general equation of a parabola is: $(y - k)^2 = 4A(x - h)$, where h and k are the co-ordinates of the vertex. The vertical (or horizontal) distance from the vertex to the focus, and from the vertex to the directrix is A. The focus lies within the parabola, the directrix is a line that lies outside the parabola

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Exercise 1

Graphical Representation of Properties

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1) Factorize and hence solve the following quadratic equations

a) $x^2 = 0$

b) $x^2 - 4 = 0$

c) $x^2 + x - 6 = 0$

d) $x^2 - 6x + 9 = 0$

e) $x^2 - 4x + 3 = 0$

f) $-x^2 - 5x - 6 = 0$

g) $2x^2 + 8x + 8 = 0$

h) $3x^2 - x - 10 = 0$

i) $x^2 + x + 8 = 0$

j) $4x^2 + 4x + 1 = 0$

k) $x^2 + 2x + 3 = 0$

2) Complete the square and hence identify the turning point of the following functions

a) $y = x^2$

b) $y = x^2 - 4$

c) $y = x^2 + x - 6$

d) $y = x^2 - 6x + 9$

e) $y = x^2 - 4x + 3$

f) $y = -x^2 - 5x - 6$

g) $y = 2x^2 + 8x + 8$

h) $y = 3x^2 - x - 10$

i) $y = x^2 + x + 8$

j) $y = 4x^2 + 4x + 1$

k) $y = x^2 + 2x + 3$

3) Using your answers to questions 1 and 2, graph the following functions

a) $y = x^2$

b) $y = x^2 - 4$

c) $y = x^2 + x - 6$

d) $y = x^2 - 6x + 9$

e) $y = x^2 - 4x + 3$

f) $y = -x^2 - 5x - 6$

g) $y = 2x^2 + 8x + 8$

h) $y = 3x^2 - x - 10$

i) $y = x^2 + x + 8$

j) $y = 4x^2 + 4x + 1$

k) $y = x^2 + 2x + 3$

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4) Using your graphs from question 3, what value(s) of x (if any) make the following inequalities true?

a) $x^2 \leq 0$

b) $x^2 - 4 < 0$

c) $x^2 + x - 6 > 0$

d) $x^2 - 6x + 9 < 0$

e) $x^2 - 4x + 3 < 0$

f) $-x^2 - 5x - 6 \geq 0$

g) $2x^2 + 8x + 8 < 0$

h) $3x^2 - x - 10 > 0$

i) $x^2 + x + 8 < 0$

j) $4x^2 + 4x + 1 > 0$

k) $x^2 + 2x + 3 > 0$

5)

a) From your previous answers, what is the relationship between the solutions to a quadratic equation and the point(s) where the graph of the equation intersects the x axis?

b) From your previous answers, what is the relationship between the solutions to an inequality and the graph of the equation?

6) By graphing the quadratic equations determine which values of x makes the following inequalities true

a) $x^2 + 1 \leq 0$

b) $x^2 + 3x < -2$

c) $x^2 - 5x + 7 > 3$

d) $x^2 - 2x - 8 < 12$

e) $x^2 + x - 17 > 5$

f) $x^2 + 2x + 3 < 2$

g) $x^2 - x + 8 > 2$

h) $-x^2 - 12x + 10 > -10$

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Exercise 2

Identities & Determinants

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1) Calculate the determinant of the following quadratic functions, and hence determine how many solutions exist for each

a) $y = x^2 - 3x + 2$

b) $y = 2x^2 + 4x - \frac{3}{2}$

c) $y = x^2 + 6x - 9$

d) $y = 3x^2 + 3x + 1$

e) $y = 4x^2 - 8x + 4$

f) $y = 3x^2 + 5x - \frac{5}{4}$

g) $x^2 - \frac{4}{3}x - \frac{1}{3}$

h) $x^2 - \sqrt{2}x + 1$

i) $-2x^2 - 6x - 5$

2) Express each of the following in the form $Ax(x-1) + Bx + C$; where: $A = a, C = c, and\ B = (a + b)$

a) $y = x^2 + 5x + 6$

b) $y = x^2 - 2x + 8$

c) $y = x^2 - 2$

d) $y = 2x^2 - 3x + 6$

e) $y = 4x^2 + 3x - 5$

f) $y = x^2$

g) $y = \frac{1}{2}x^2 + \frac{1}{3}x - \frac{1}{4}$

h) $y = -x^2 - x - 1$

i) $y = -3x^2 - 3x - 3$

3) Find the quadratic equation that fits each of the three sets of points below

a) (1,2) (0,6) (3,0)

b) (2,8) (1,5) (-1,5)

c) (1,3) (-2,18) (-1,9)

d) (2,-2) (-1,9) (0,6)

e) (1,1) (-2,-8) (-1,1)

f) ($\frac{1}{2}$,-1) (1,0) (2,6)

g) (2,4) ($\frac{3}{2}$,$\frac{9}{4}$) (-3,9)

h) (1,2) (-2,20) (0,2)

i) (1,-5) (2,7) ($\frac{1}{2}$, -8)

j) (1,64) (-1,4) ($\frac{1}{3}$, 36)

4) Solve the following by first reducing them to quadratic equations of the form
$ax^2 + bx + c = 0$

a) $x^4 + x^2 - 6 = 0$

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b) $x^6 - 4x^3 + 4 = 0$

c) $4x^4 + 2x^2 - 8 = 0$

d) $8x + 4\sqrt{x} - 1 = 0$

e) $(x + 2)^2 = 4x^2 + 1$

f) $(x - 3)^2 + 2 = (x + 1)^2 - 1$

g) $(x - 4)^2 - 12 = x + 1$

h) $4^x - 2(2)^x + 1 = 0$

i) $16^x - 5(4)^x + 6 = 0$

j) $81^x - 4(3)^{2x} + 3 = 0$

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Exercise 3

Equations of Parabolas

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1) Find the equations of the parabolas defined by the given focus, axis and directrix.

a) Focus at (0,1), axis $x = 0$, directrix $y = -1$

b) Focus at $(0,\frac{1}{2})$, axis $x = 0$, directrix $y = -\frac{1}{2}$

c) Focus at $(0,\frac{1}{4})$, axis $x = 0$, directrix $y = -\frac{1}{4}$

d) Focus at (0,4), axis $x = 0$, directrix $y = -4$

2) Find the equations of the parabolas defined by the given focus, axis and directrix.

a) Focus at (2,1), axis $x = 2$, directrix $y = -1$

b) Focus at (3,-3), axis $x = 3$, directrix $y = 3$

c) Focus at (-2,-2), axis $x = -2$, directrix $y = 2$

d) Focus at $(1,\frac{1}{2})$, axis $x = 1$, directrix $y = -\frac{1}{2}$

3) Find the equations of the parabolas defined by the given focus, axis and directrix.

a) Focus at (0,-4), axis $x = 0$, directrix $y = 6$

b) Focus at (0,-2), axis $x = 0$, directrix $y = 2$

c) Focus at (0,1), axis $x = 0$, directrix $y = -3$

d) Focus at (0,3), axis $x = 0$, directrix $y = -1$

4) A Find the equations of the parabolas defined by the given focus, axis and directrix.

a) Focus at (3,1), axis $x = 3$, directrix $y = 0$

b) Focus at (2,-4), axis $x = 2$, directrix $y = -6$

c) Focus at $(1,\frac{1}{2})$, axis $x = 1$, directrix $y = 1$

d) Focus at (-2,-1), axis $x = -2$, directrix $y = 5$

5) By rewriting the following in parabolic form, find the focus, vertex, axis and directrix

a) $y = x^2$

b) $y = x^2 + 4$

c) $y = x^2 - 3x + 2$

d) $y = 2x^2 + 3x - 2$

e) $y = \frac{1}{2}x^2 - \frac{1}{4}x + 1$

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f) $y = 4x^2 - 6x + 2$

6) Find the general equation of the parabola with axis $x = 2$, and vertex at the point $(2, y_0)$ by considering the values of $y_0$to be

a) -1

b) -4

c) 1

d) 0

e) 3

f) $\frac{1}{2}$

7) Find the general equation of the parabola with axis $x = -3$, having a focal length of A by considering the values of A to be

a) 2

b) 4

c) 1

d) -3

e) 0

f) -2

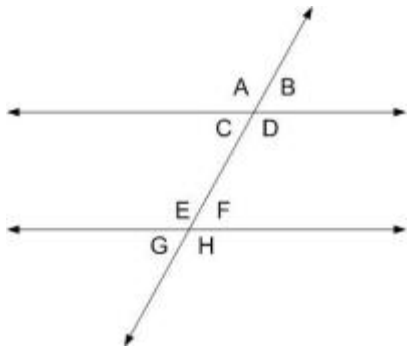
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Year 11 Unit 2 Mathematics

Plane Geometry

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Useful formulae and hints



- C and F are alternate interior angles; they are equal
- A and H are alternate exterior angles; they are equal
- A and E are corresponding angles; they are equal
- A and B are adjacent angles; they total 180°
- B and C are vertically opposite angles; they are equal
- C and E are co-interior angles; they total 180°

- The sum of the interior angles of a triangle is 180°
- Tests for similar triangles
    - AAA
    - SSS
    - SAS
- Tests for congruent triangles
    - SSS
    - SAS
    - ASA
    - AAS
    - Hypotenuse, side
- Pythagoras' Theorem:$a^2 + b^2 = c^2$, where c is the hypotenuse
- Areas
    - Triangle: $\frac{1}{2} \times$ base $\times$ perpendicular height
    - Rectangle: length x breadth

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- Parallelogram: Length × perpendicular height
- Trapezium: $\frac{a+b}{2}$ × height, where a and b are the two parallel sides

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Exercise 1

Angles Formed by Transversals

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1) From the diagram below, give examples of the following pairs of angles

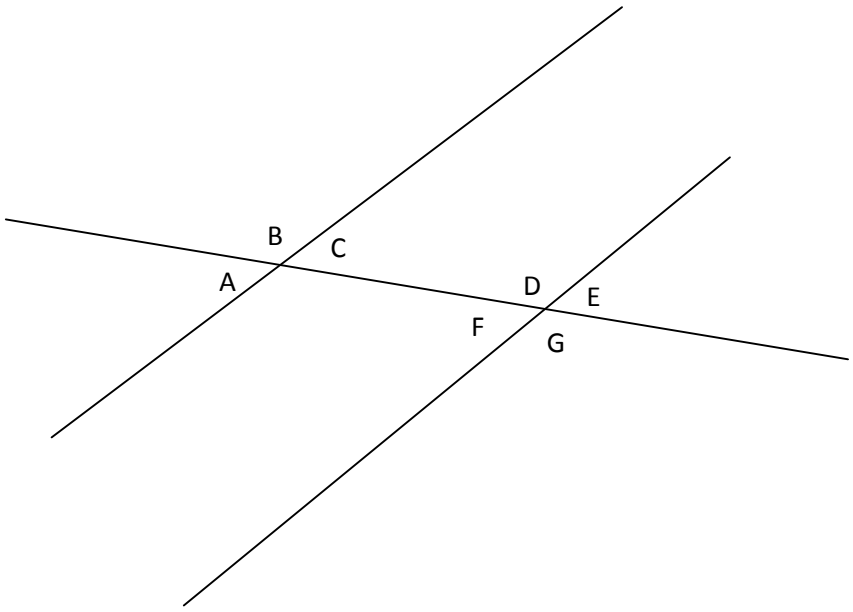
a) Vertically opposite

b) Alternate interior

c) Corresponding

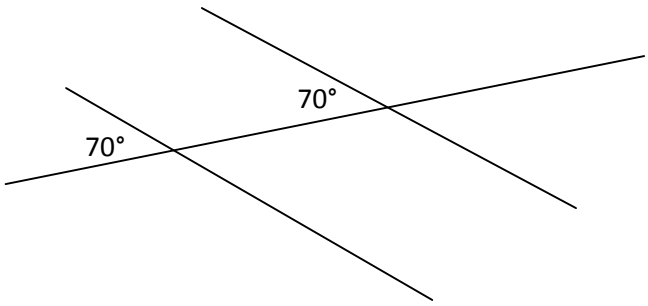
d) Co-interior

e) Alternate exterior



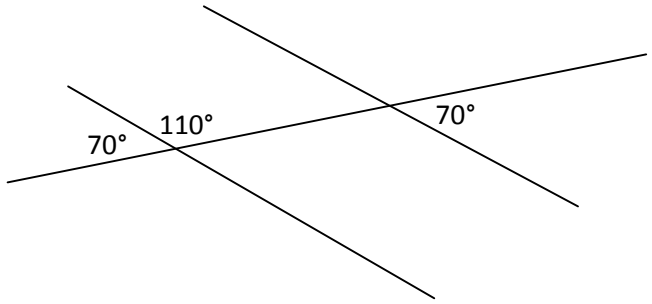
2) Identify which diagrams show parallel and which show non parallel lines; give reasons for your answers

a)

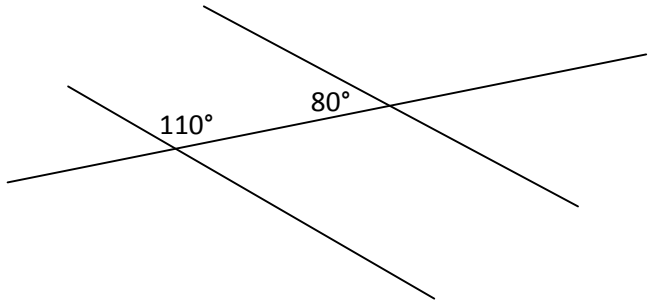


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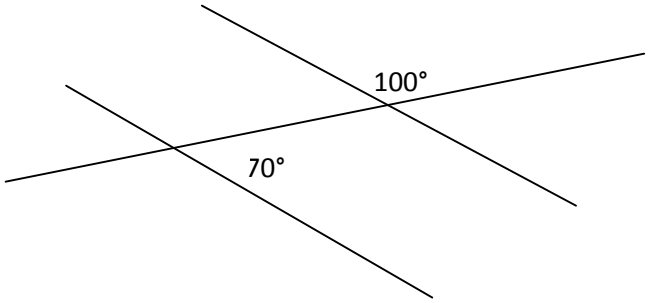
b)



c)

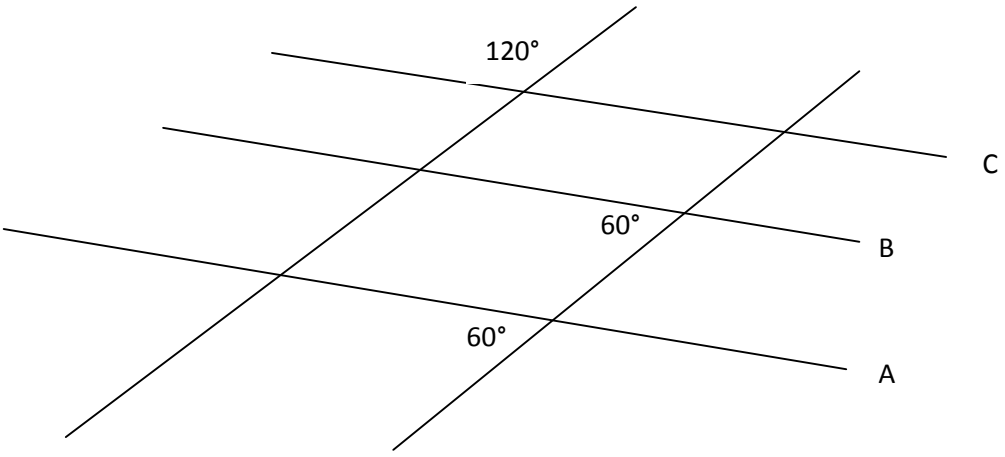


d)



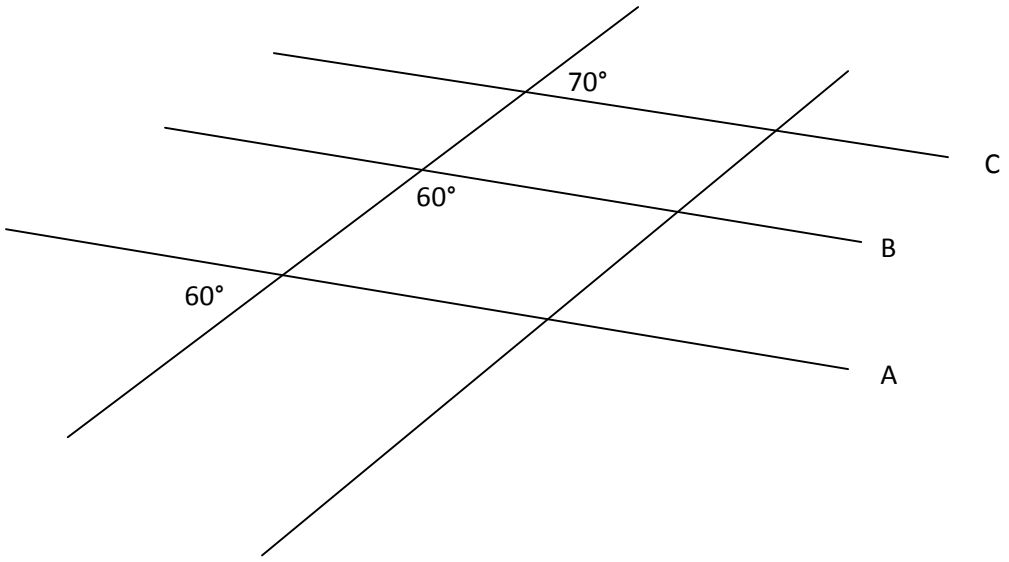
3) For each of the diagrams below, state which of the lines A, B and C are parallel to each other, giving reasons for your answers. Assume that the transversals are parallel to each other

a)

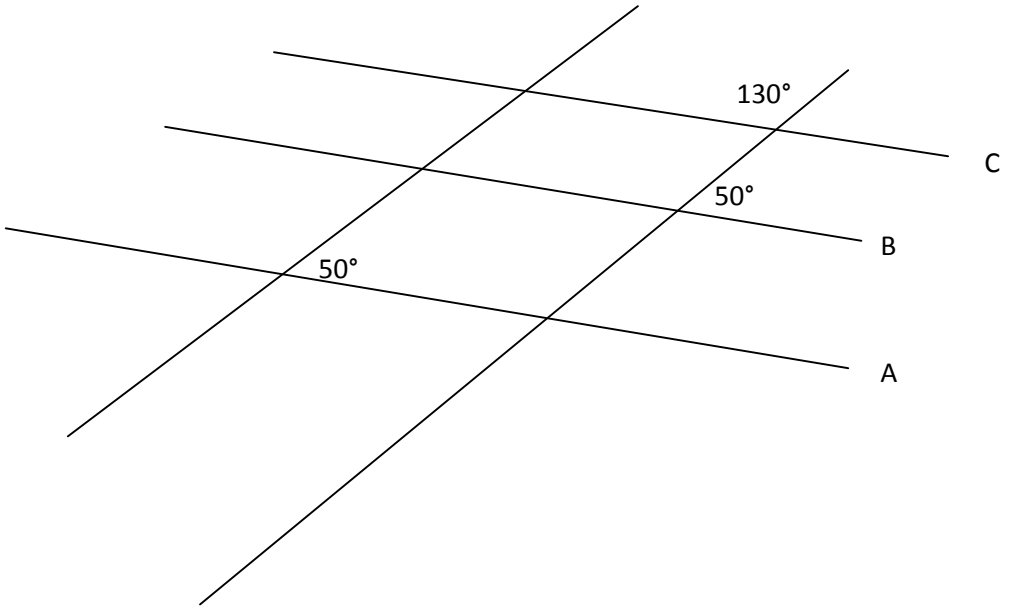


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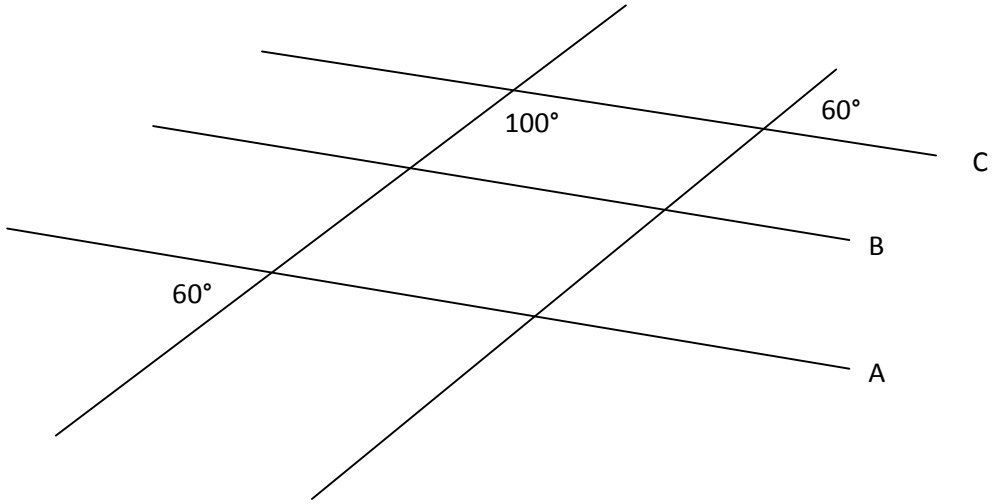
b)



c)



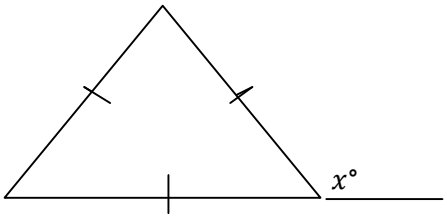
d)



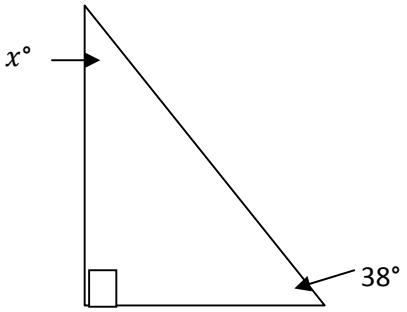
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4) Find the value of x in each of the following

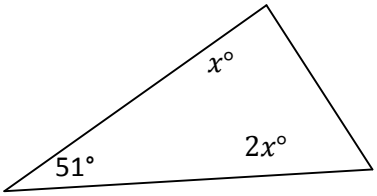
a)



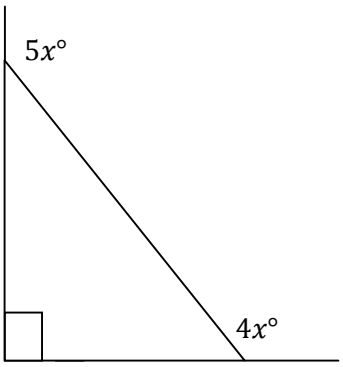
b)



c)

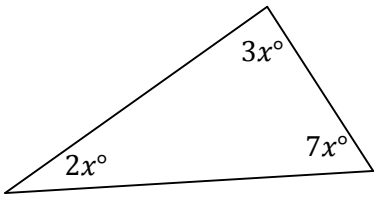


d)

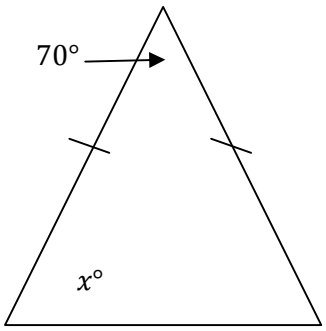


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e)



f)



5)

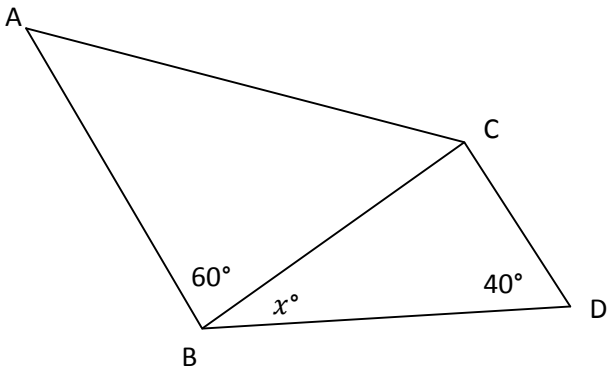
a) Find the size of an interior angle of a regular pentagon

b) What is the sum of the internal angles of a regular octagon?

c) What is the sum of the external angles of a regular nonagon (Taking one angle per vertex only)?

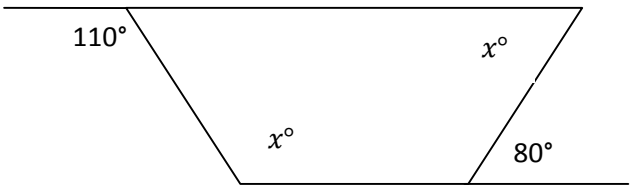
6) Find the value of x in the following

a) AB || CD

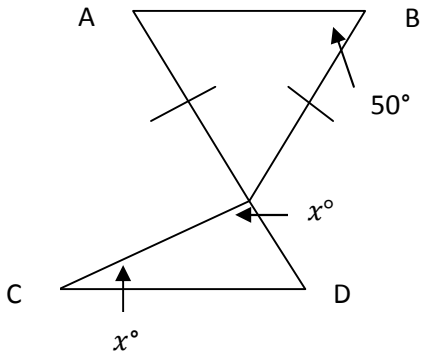


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b)



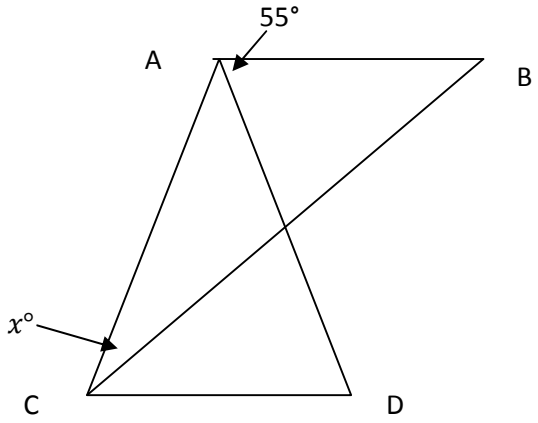
c) AB || CD



d) AB || CD

AD $\perp$ BC

AD = AC



Find the size of angle ACB

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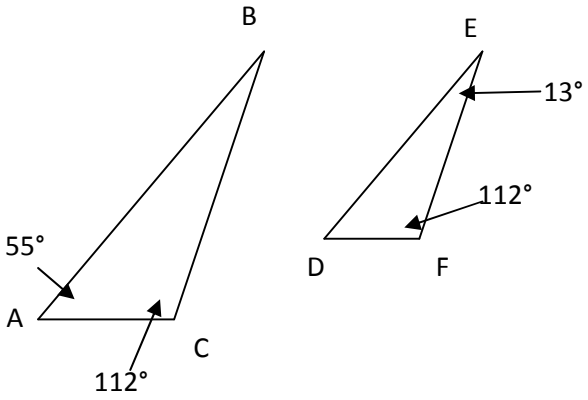
Exercise 2

Similarity & Congruence

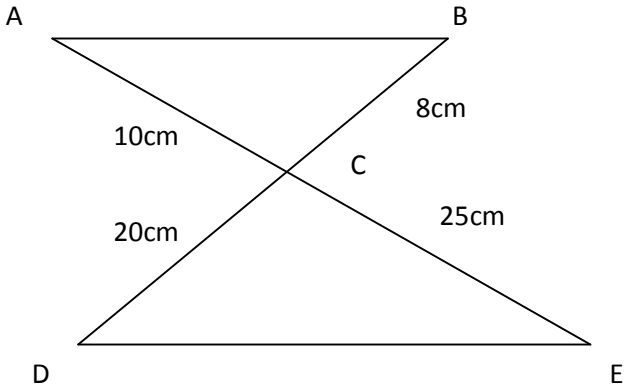
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1) Determine if each pair of triangles is similar. If so, state the similarity conditions met

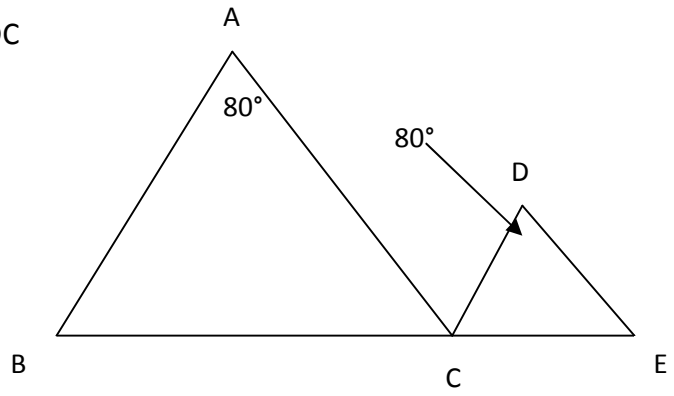
a)



b)

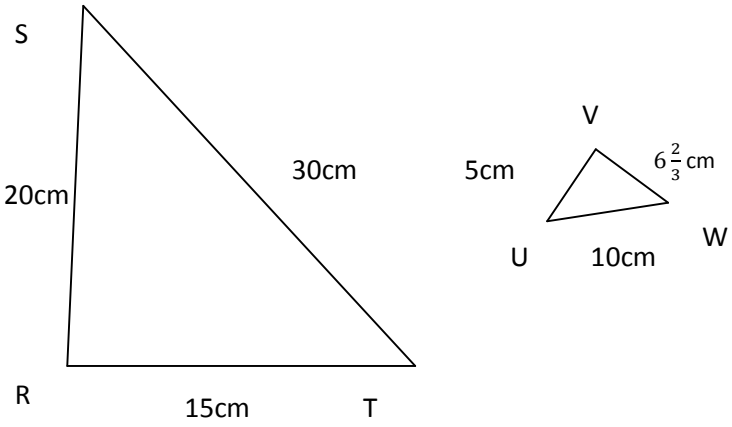


c) AB || DC

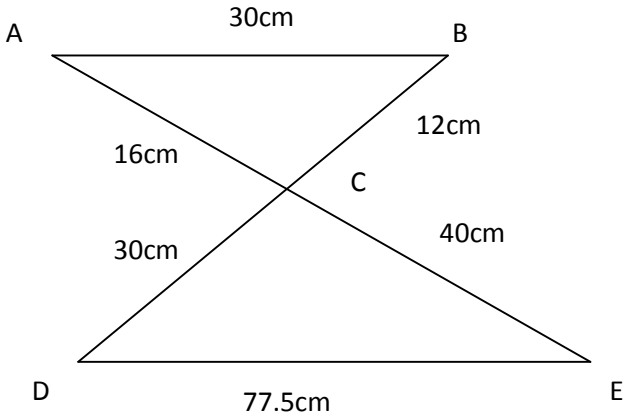


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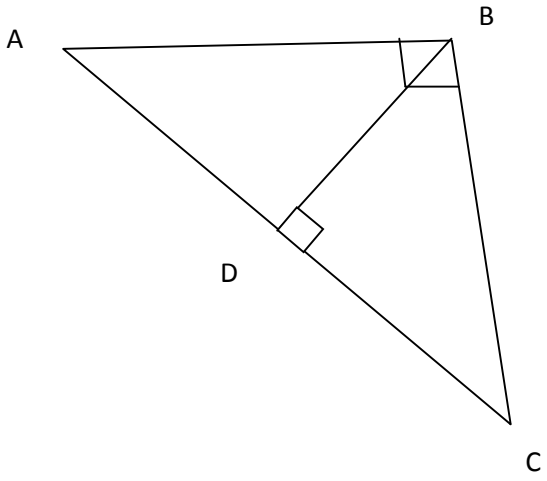
d)



e)



f)



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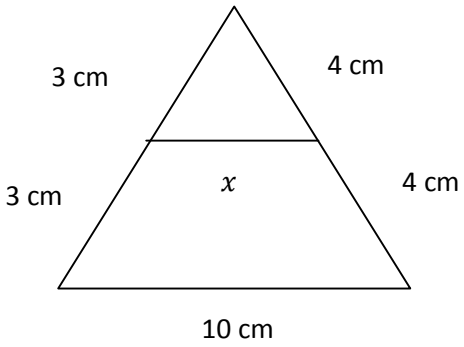
2) A tower casts a shadow of 40 metres, whilst a 4 metre pole nearby casts a shadow of 32 metres. How tall is the tower?

3) A pole casts a 4 metre shadow, whilst a man standing near the pole casts a shadow of 0.5 metres. If the man is 2 metres tall, how tall is the pole?

4) A ladder of length 1.2 metres reaches 4 metres up a wall when placed on a safe angle on the ground. How long should a ladder be if it needs to reach 10 metres up the wall, and be placed on the same safe angle?

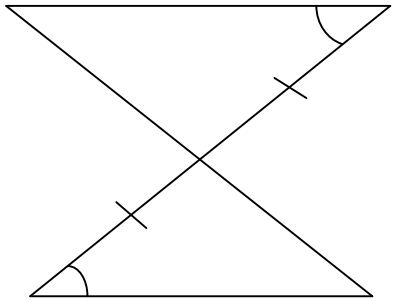
5) A man stands 2.5 metres away from a camera lens, and the film is 1.25 centimetres from the lens (the film is behind the lens). If the man is 2 metres tall how tall is his image on the film?

6) What is the value of x in the following diagram?



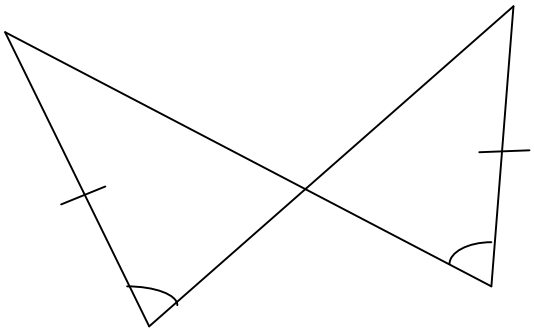
7) State which of the following pairs of triangles are congruent, and the reasons for their congruency

a)

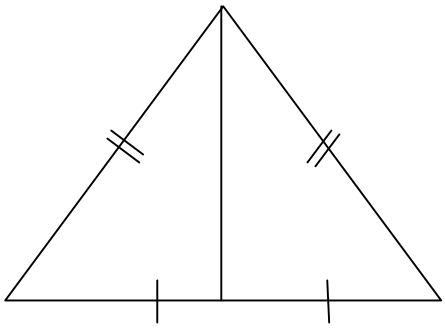


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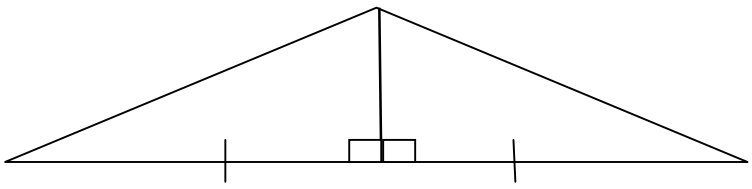
b)



c)

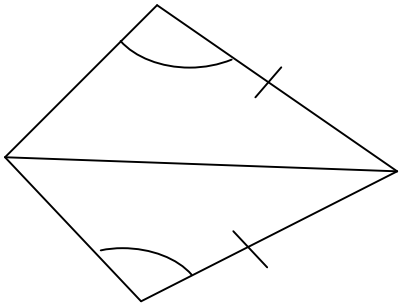


d)

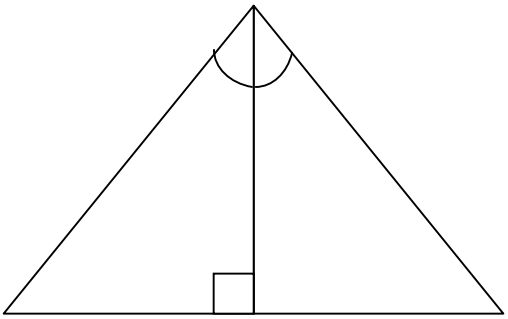


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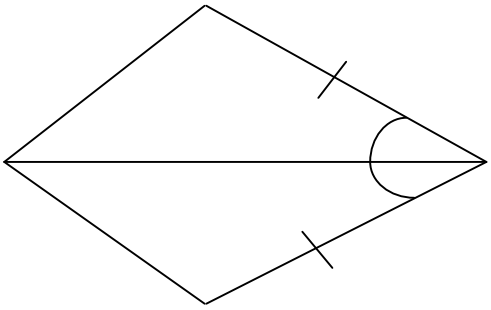
e)



f)



g)



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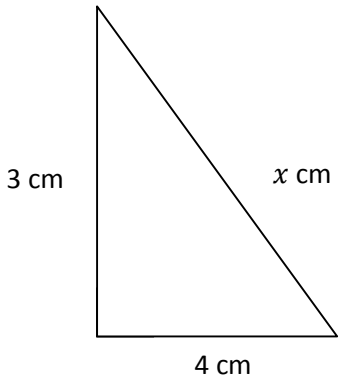
Exercise 3

Pythagoras' Theorem

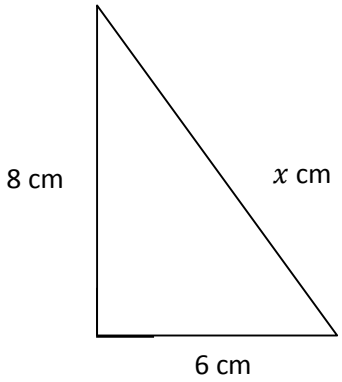
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1) Find the value of x to 2 decimal places in the following diagrams

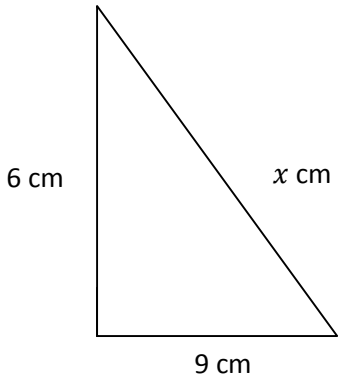
a)



b)

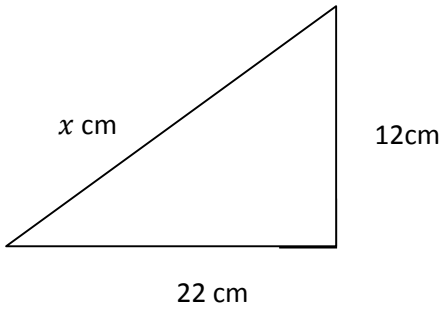


c)

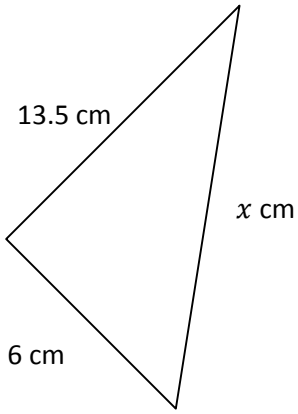


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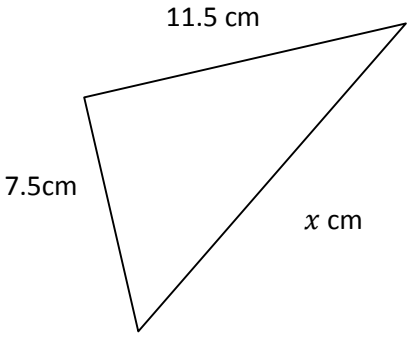
d)



e)



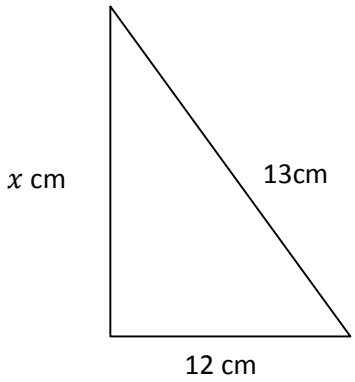
f)



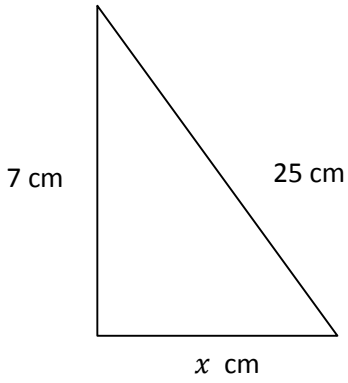
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2) Find the value of x to 2 decimal places in the following diagrams

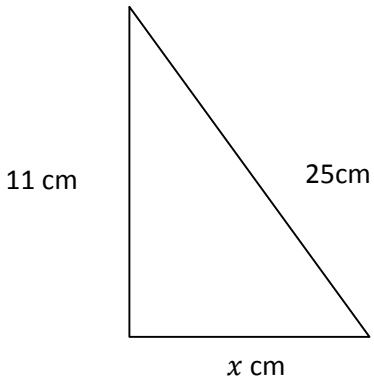
a)



b)

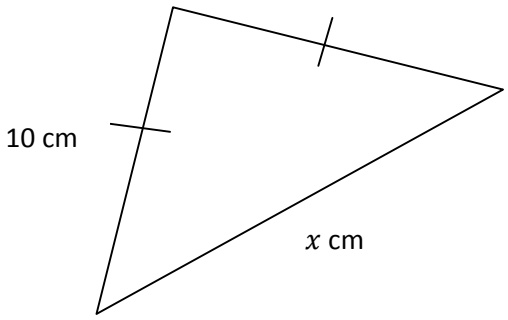


c)

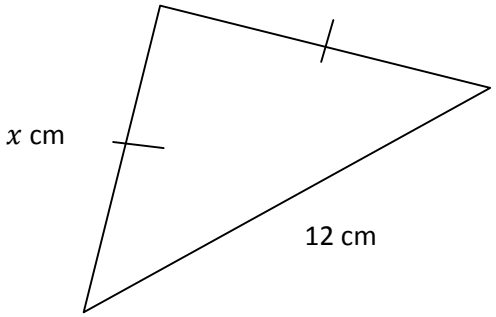


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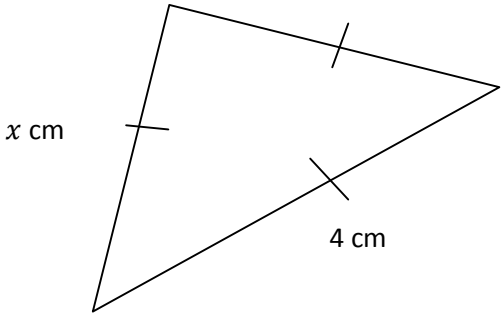
d)



e)



f)



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3) A man walks 5 km east then turns and walks 8 km south. How far is the shortest distance to his starting position?

4) A ladder 2 meters long is placed against a wall and reaches 1.5 meters up the wall. How far is the foot of the ladder from the base of the wall?

5) A farmer wishes to place a brace across the diagonal of a rectangular gate that is 1.8 metres long and 0.6 metres wide. How long will the brace be?

6) A square room measures 11.7 metres from corner to corner. How wide is it?

7) The size of television sets are stated in terms of the diagonal distance across the screen. If the screen of a set is 40 cm long and 30 cm wide, how should it be advertised?

8) A student has two choices when walking to school. From point A, he can walk 400 metres, then turn 90° and walk a further 200 metres to point B (school), or he can walk across the field that runs directly from A to B. How much further does he have to walk if he takes the path instead of the field?

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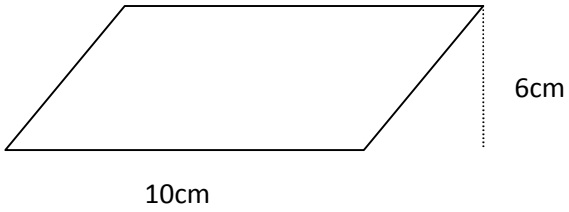
Exercise 4

Area Calculations

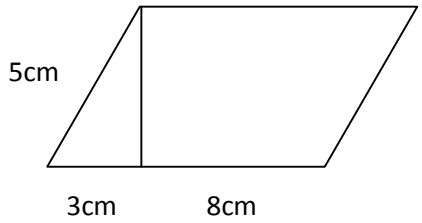
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1) Find the area of the following

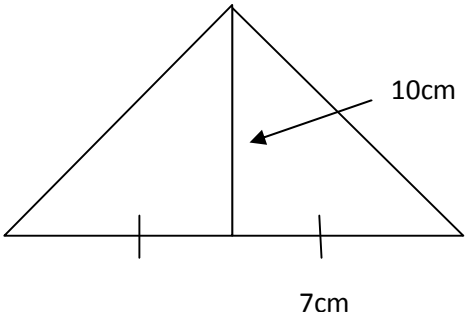
a)



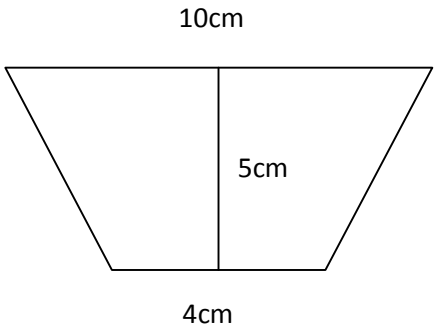
b)



c)



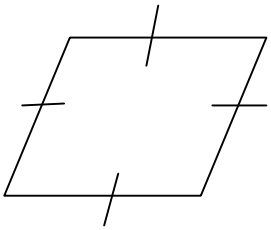
d)



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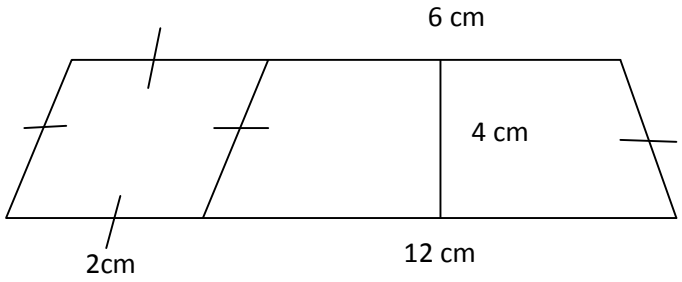
e) Perimeter = 12 cm

Perpendicular height = 4cm

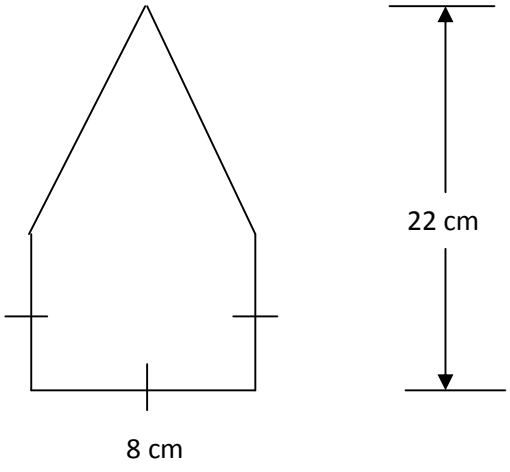


2) Calculate the area of the following composite shapes

a)

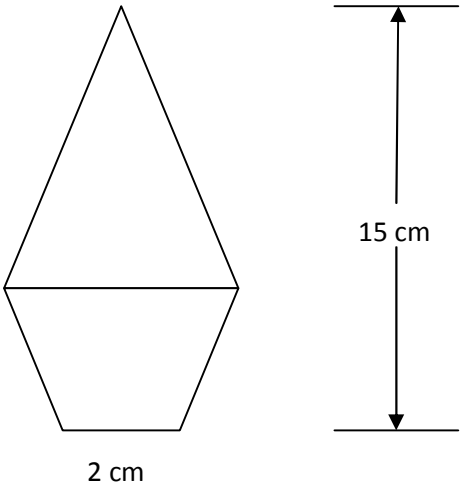


b)

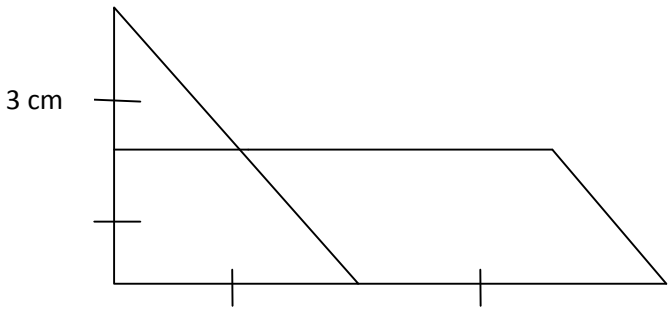


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c) Area of triangle = 40 cm^2



d)



3) A badge is in the shape of an equilateral triangle with a perimeter of 18cm. What is the area of the badge?

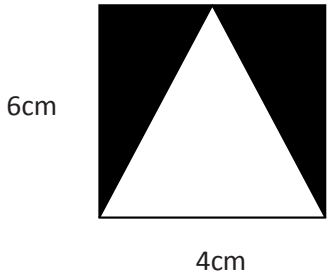
4) A rhombus has one diagonal measuring 8cm and the other measuring 6cm. What is its area?

5) What height must an isosceles triangle of base 2cm be in order to have an area the same as an equilateral triangle of side length 4cm?

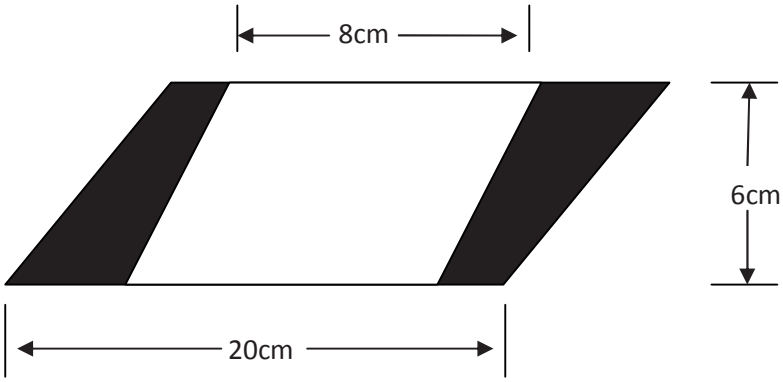
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6) Calculate the area of the shaded regions

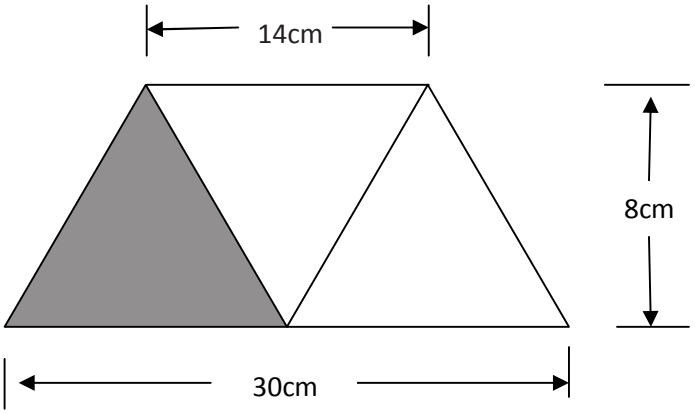
a)



b)

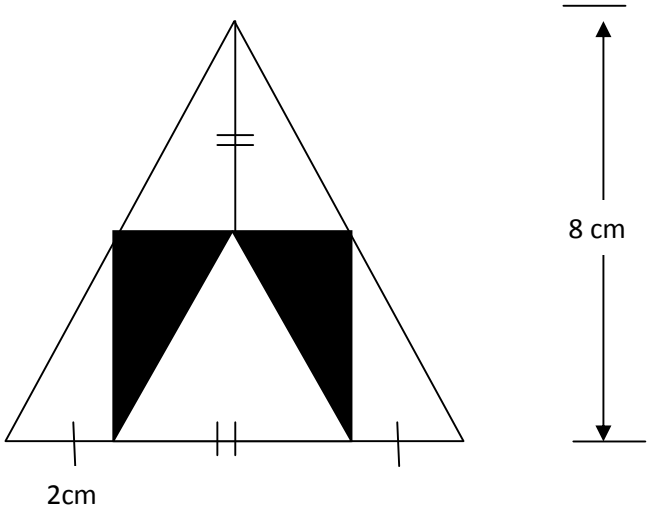


c)



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d) Area of large triangle = 32 cm^2



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Year 11 Unit 2 Mathematics

Derivative of a Function

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Useful formulae and hints

- A function f is continuous at a point **a** if the following conditions are satisfied.
  - f(a) is defined
  - $\lim_{x\to a} f(x)$ exists
  - $\lim_{x\to a} f(x) = f(a)$
- If $f(x) = x^n, then\ f'(x) = nx^{n-1}$
- If $f(x) = g(x)h(x), f'(x) = h(x)g'(x) + g(x)h'(x)$
- $f\big(g(x)\big) = g'(x) \times f'\big(g(x)\big)$:
  Example: $f(x) = (x^2 + 2)^2$
  $f'(x) = 2x \times 2(x^2 + 2)$

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Exercise 1

Continuity

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1) Graph the following functions in the domain $-3 \leq x \leq 3$

a) $f(x) = x^2$

b) $f(x) = 2x + 3$

c) $f(x) = \frac{1}{1+x}$

d) $f(x) = \frac{1}{\sqrt{x}}$

e) $f(x) = \frac{x^2-1}{x+1}$

f) $f(x) = \frac{1}{2-x}$

2) Using your graphs in question 1 as a guide, state whether functions are continuous or discontinuous over the domain. Give mathematical proof

3) Show at what point(s) the following functions are discontinuous

a) $f(x) = \frac{2}{x^2-x}$

b) $f(x) = \frac{x+1}{x+3}$

c) $f(x) = \begin{bmatrix} x+2 & for\ x < 0 \\ \frac{3}{x^2+4} & for\ x \geq 0 \end{bmatrix}$

d) $f(x) = \frac{1}{x^2-1}$

e) $f(x) = \frac{|x|}{x}$

4) Let $f(x) = x^2, g(x) = \frac{1}{1-x}, h(x) = \frac{\cos x}{x}, r(x) = x^3 - 1$
State whether the following functions are continuous, and give reasons

a) $k(x) = x^2 + \frac{1}{1-x}$

b) $k(x) = x^2 - \frac{\cos x}{x}$

c) $k(x) = x^2(x^3 - 1)$

d) $k(x) = \frac{1}{1-x} + \frac{\cos x}{x}$

e) $k(x) = \frac{1}{1-x} - x^2$

f) $k(x) = \frac{x^3-1}{1-x}$

g) $k(x) = \frac{\cos x}{x} + (x^3 - 1)$

h) $k(x) = \frac{x^2(\cos x)}{x}$

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Exercise 2

Secant to a Curve

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1) Using the curve $y = x^2$, determine the gradient of the line joining the following points on the curve(the secant)

a) (-4,16) and (-2,4)

b) (0,0) and (-1,1)

c) (2,4) and 5,25)

d) (2,4) and (-2,4)

e) $\left(\frac{1}{2},\frac{1}{4}\right)$ and $\left(-\frac{1}{4},\frac{1}{16}\right)$

2) For the same curve, determine the gradient of the secant from the point (1,1) to the following points

a) (-4,16)

b) (-3,9)

c) (-2,4)

d) (-1,1)

e) (0,0)

f) $\left(\frac{1}{2},\frac{1}{4}\right)$

g) $\left(\frac{3}{2},\frac{9}{4}\right)$

h) (2,4)

i) (3,9)

j) (4,16)

k) (5,25)

3)

a) Does the pattern of numbers in question 2 suggest that there is a limiting value for the gradient of the secant to the point (1, 1) as$x \to 1$? If so what is that value?

b) What is the general equation for the limit of the gradient of the secant to the point (1, 1) as$x \to 1$?

c) Calculate the limit of the gradient of the secant to the point (1, 1) as $x \to 1$

4) Calculate and hence construct a table of the limits of the gradient of the secant to the function $f(x) = x^2$ at the following points

a) (-1,1)

b) (2,4)

c) (-4,16)

d) (3,9)

e) (10,100)

5) Formulate a rule for the value of the gradient of the secant to the curve $f(x) = x^2$ at any point

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Exercise 3

Methods of Differentiation

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1) Using the equation $f'(c) = \lim_{h\to 0} \frac{f(c+h)-f(c)}{h}$ calculate the derivative of the functions for the following values of x.

a) $f(x) = x^2$ at $x = 1, x = -2$

b) $f(x) = x^2 + 3$ at $x = 2, x = 1$

c) $f(x) = x^3 - 3x$ at $x = -1, x = 3$

d) $f(x) = x^2 + 2x + 4$ at $x = 1, x = -1$

e) $f(x) = x^3 - 6x$ at $x = 0, x = 2$

f) $f(x) = x^3 + 2x + 1$ at $x = 1, x = 2$

2) From question 1, find the equation of the tangent line to each equation at the specified points

3) Graph each of the functions from question 1 and their derivatives (use the same graph for each function and derivative)

4) Find $\frac{dy}{dx}$ for each of the following functions

a) $y = x^2$

b) $y = x^2 - 2x + 12$

c) $y = 2x^2 - 2x$

d) $y = \frac{1}{2}x^2 - x$

e) $y = x^4 + 3x^3 - 4x^2 + \frac{1}{2}x + 2$

5) Find $\frac{d}{dx}(f(x))$, where $f(x) =$

a) 4

b) $2x$

c) $\frac{1}{3}x^3 - 4x$

d) $2x^5 - 4x^2$

e) $\frac{1}{4}x^4 + \frac{1}{3}x^3 - \frac{1}{4}x^2 + 100$

6) Find the derivative of the following functions

a) $f(x) = \sqrt{x}$

b) $f(x) = x^{-1}$

c) $f(p) = \frac{1}{p^2}$

d) $f(t) = \frac{1}{\sqrt{t}}$

e) $f(a) = \sqrt{a} + \frac{1}{a}$

f) $f(x) = \sqrt[3]{x} - x^{-1}$

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7) Find $f'(x)$ using the product rule, where $f(x) =$

a) $x^2(\sqrt{x})$

b) $\frac{1}{x}(2x-3)$

c) $\frac{1}{x^2}(x^2-4)$

d) $2x^{-1}\left(\frac{1}{2}x^3\right)$

e) $\frac{3x}{2}(2x^2-1)$

f) $\frac{x^2}{4}\left(x^2-\frac{1}{4}x\right)$

8) Find the derivatives of the following functions

a) $f(x) = (x^2-2)^2$

b) $f(t) = (t^2+1)^{-1}$

c) $f(a) = \frac{a+1}{a-3}$

d) $g(x) = \frac{x^3-4x+2}{x}$

e) $h(z) = \frac{1}{(x^2-1)^2}$

f) $f(x) = \frac{x^2}{x-2}$

g) $f(x) = \sqrt{x^2-3}$

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